# OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

# OpenSourceRail v0.5.2 development — fixed-zone ferrite maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.

Low CAPEX, OPEX and long-term debt repayment are central design objectives. The detailed city briefs restore line-by-line service, energy, manufacture and whole-life affordability views for the PM-maglev station-launched monorail. Each current city has low/reference/high scope-covering capital budgets, annual operating cash and debt schedules, with explicit unquoted assumptions. They distinguish resource cost from borrower cash and imported inputs from foreign-currency debt. Complete engineered costs and a proven comparative advantage remain open; no former railway savings percentage is inherited.

Station-led GOA4 dispatch concentrates launch and receiving control at known sites. The following train may use a released berth/resource only after valid full-tail clearance and protected authority; a launch command alone cannot release occupancy. Independent active support, separation, safe arrival, passenger protection and recovery remain necessary. The city defaults retain exclusive whole-leg headways while the progressive multi-train candidate undergoes separate qualification.




OpenSourceRail maglev station concept

Illustrative development concept, without physical or manufacturing acceptance.

Fixed linear synchronous motors launch and receive the train at stations. Selected longer legs add located midway acceleration; remaining guideway is unpowered; onboard ferrite PM-biased active support, guidance and passenger services remain independently powered. Removing main traction equipment from the train shifts electrical plant to stations. Support coils/drives, cooling, sensing, protection, doors, HVAC and emergency equipment remain necessary. Contactless running aims to reduce contact noise and wear; actual losses, acoustic performance and maintenance require measurement.

## Current detailed candidate

The new detailed-system chain accounts 25.619t tare/37.619t AW3 per car for one dimensioned candidate. Actual stock, current circuits, loads, manufacturing quantities and native identities agree on its revision. Corrected thin return-rail saturation prevents requiredAW3 support; propulsion demagnetisation/convergence failures and structural/protection holds remain; no complete engineered or adopted train is claimed. Earlier targets and actuator passes below retain their original configuration.

The owner/builder planner restores city edit proposals,16GIS layers and generic delivery workfront/procurement/cash plans. Independent SUMO, IFC/IDS/BCF, typed field negative controls and native business navigation have explicit bounded scopes.

## Body, civil works and local capability

Baghdad uses six 17 m bodies. The central tare target is 14.2 t/car, including a 5.2 t equipped structural GFRP body, 3 t ferrite, 30 kWh services storage and an existing 1.8 t growth reserve. Four passenger doors per car are the body takeoff. Counted frame families, supported panels, couplers and hardpoint candidates now have calculated dimensions and margins; frame/emergency-brake growth consumes this reserve. Joined-body strength, complete equipment, fatigue, CG and inertia remain unqualified. The50kWh wider-corridor reference is 14.4t tare/ 26.4t AW3 per car. Its pack increase is included once; the primary detail/emergency screens use that heavier mass alongside the retained30kWh case.

The current ordinary civil candidate is a 25 m, 0.80 × 2.50 m closed concrete box with 100 mm webs, 150 mm flanges and end infills. The supplied 1.80 m-depth case remains a separate comparison. Hollow columns/caps and alternative foundation families are compared by quantity. Necessary passive iron, anchors, bearings, connections and final survey remain in scope. Ground-specific selection, prestress/handling details, fixed/guided bearings and emergency load paths have numerical screens; physical structural and construction acceptance remain open.

The research target omits a continuous external walkway. Vertical spiral chutes are the preferred alternative under investigation, without walkway-removal approval. Safe landing, assisted evacuation, deployment, stopping/recovery and fire response must work at every allowed stopping location.




Vertical spiral escape chute concept

Illustrative concept; no qualified railway device or all-stop evacuation claim.

The 99% elevated-mainline objective and straight tangents aim to reach constrained corridors with less surface occupation. Rights, buildings, utilities, access, wildlife, water, curvature and foundation feasibility govern each route.

Local precast/GFRP production, steel fabrication, cartridge assembly, station/booster winding cassettes, interiors, wiring, Rust software and integration can build domestic capability. Prioritise verified British manufacture when selecting UK machinery. A UK distributor or integrator does not establish manufacturing origin. Complete local content remains unknown until the bill and input origins exist.

# Operating and cost boundaries

Urban passenger stops target at least 1.5 km spacing and about 2 km on average. Suburban stops target 4–5 km with fixed midway acceleration where required. Junctions transfer passengers between independent fixed lines, without train-routing switches. Retained close junctions remain spacing/access exceptions requiring a solution. Straight alignments are planning candidates subject to land, buildings, utilities, grades, access and evacuation.

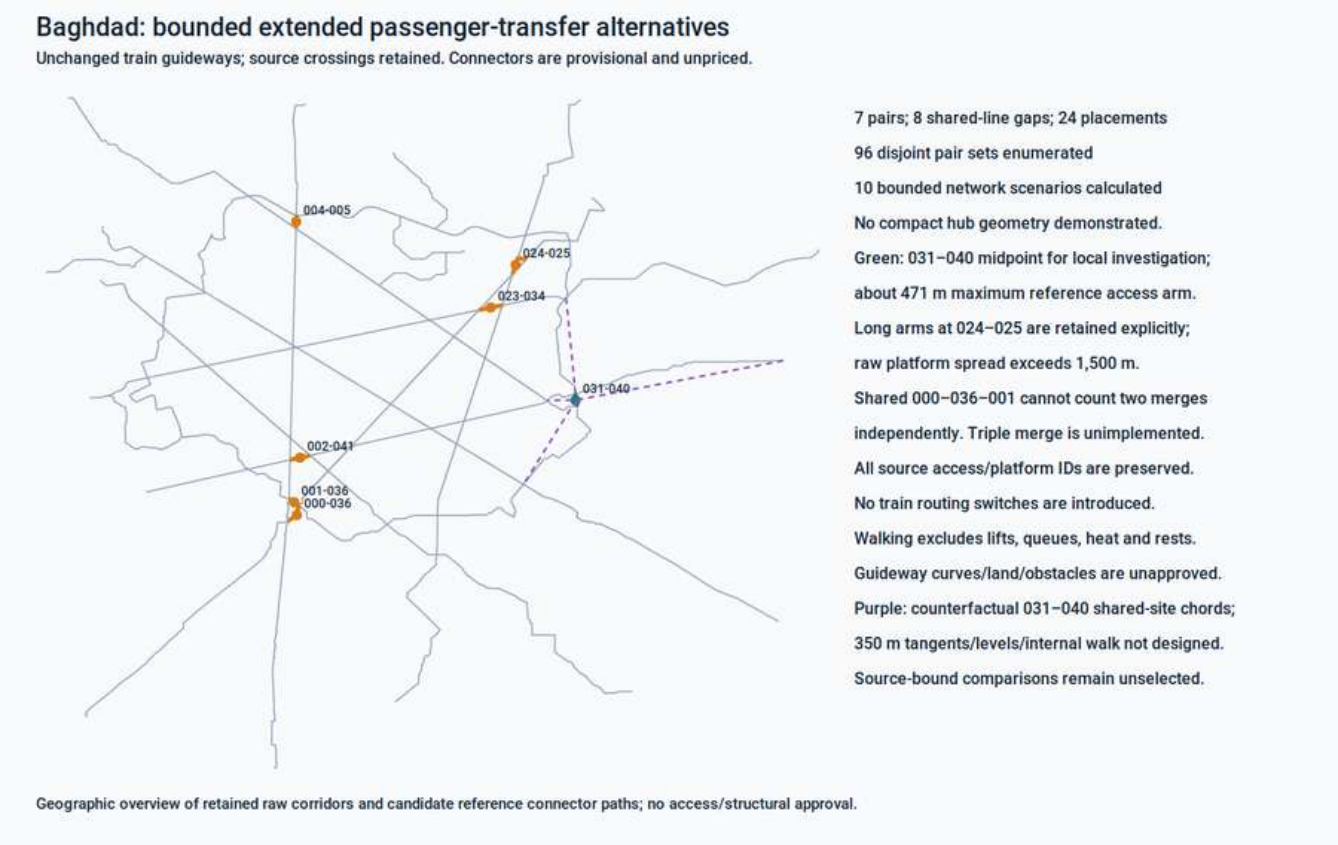
The 1 km/88 s/180 s station and moving packets remain component benchmarks. The supplied 90 s/190 s case is a separate authored comparison. Longer urban/suburban profiles have their own finite-zone, power, services, clearance and actual Rust fault tests. A separate local multi-train prototype now executes progressive Rust reservations and stopping-distance envelopes. Capacity shortfalls and unqualified route-wide sectioned protection stay visible; a timer does not prove a clear railway.

The retained benchmark uses 240 kW/car charger input at 90% assumed conversion efficiency. Longer cases compare their own pack, charger and dwell choices. Independently equipped rescue nodes, faster winches and 30/50/60 kWh packs have explicit rope/drum/portal/foundation, drive/power, mass and cost comparisons. A midpoint accelerator is not automatically a refuge, charging station or rescue site. Rope/termination, access, essential loads, physical hardware and real ground remain unqualified.

City subtotals separately count eight scopes: fleet, ordinary and terminal civil work, station electrical targets, a conditional footing increment, booster electrical plant, booster mount stock and independent rescue kits. Complete station/transfer hubs, grid reinforcement, land, factory capital, evacuation, qualification, finance and other unpriced work remain additional. Fewer passenger platforms do not by themselves establish a total saving.

# Baghdad interchange investigation

The default Baghdad plan remains 273 proposed stops with 10 transfer-spacing conflicts. A separate interchange investigation compares extended walking links on unchanged guideways with coordinated shared-site hub concepts that change guideway geometry. No alternative is selected.




Retained Baghdad corridors, extended transfer links and unselected shared-site concept

Original boarding/access records remain traceable. The extended links retain reference walks and partial electrical target credits, without complete connector costs or ridership-weighted benefits. Shared-site concepts draw involved fixed lines through one location; the original100m compact walk is only a sensitivity. The separate geometry study below fits new approaches, powered zones, platforms and actual internal paths without selecting or qualifying a hub. Lift resilience, peak transfer pulses, shade, resting places, evacuation, land and complete construction/maintenance costs must enter selection. Trains never switch between lines in these passenger-transfer proposals.

The separate access engineering study adds physical connector/support quantities, accessible routes, passenger pulses and lift-outage comparisons. Assumed loads, flow rates and geometry remain provisional; no option is selected and no complete construction price is claimed.

The shared-hub geometry study develops new station tangents, both directional powered zones, bounded approach transitions, platforms and actual door-to-door paths. Rejected fits remain visible; new quantities receive no previous civil price credit. This separate study does not turn the original100m sensitivity into a measured transfer walk.

The longitudinal-stock sweep tests failure-directed alternatives after the GFRP stiffness hold. Exact drilled steel stock and changed spacers add their mass to the complete retained AW3 basis. The unchanged phase criterion retains passes and failures; thin-wall joints, heavier loads, qualified materials and flexible-body behavior remain open. No stock alternative changes the city default.

## Requirement closure and physical evidence

The qualification programme gives each applicable requirement a deliverable, prerequisites and a route to actual test or acceptance evidence. Specimen/rig procedures and typed raw records cover materials, magnetic forces and losses, controls, civil stages, services, egress and recovery. Prepared procedures and synthetic records do not constitute physical tests. Qualified limits, exact revisions and authenticated independent decisions remain necessary.

The multi-train capacity study compares local progressive ownership with exclusive whole-leg protection and retains adverse cases. It does not promote city schedules or approve an operating railway. Portable study bundles now use bounded, checksum-verified numbered archives so that the complete retained source set can exceed one archive's size without dropping evidence.

## Inspect and reproduce the current work

| Current digital capability | Scope |
|---|---|
| City planning | 266 primary packages; located operating stops, access assignments, passenger-transfer spacing conflicts, boosters, rescue nodes, motion/headway/capacity screens and eight-part cost subtotals |
| Operating study | Separate urban/suburban profiles with finite station/booster zones, services/charging and electrical/protection fault cases |
| Interchange investigation | Unselected Baghdad extended-access variants and shared-site guideway concepts; default plan and open acceptance preserved |
| Access engineering | Connector/support quantity and passenger/lift-outage alternatives, without site or structural acceptance |
| Multi-train prototype | Actual local Rust progressive reservations, full-tail release and stopping-envelope/adverse controls; city capacity holds retained |
| Qualification programme | Exact requirement task chains, physical test protocols and guarded raw-evidence assessment; no simulated acceptance |
| Manufactured details | Counted frame/panel/hardpoint/coupler/stator interfaces, prestress/handling and emergency load-path screens, foundation cost deltas and open qualification |
| Assembled structure | Fitted native material regions, mounting holes and fasteners, retained collision controls and geometry-derived partial mass/inertia; omitted equipment remains explicit |
| Shared-hub geometry | Both directional approaches and finite powered zones, platforms, access obstacles and actual door paths; no site selection or former price credit |
| Controls | Actual Rust support replay and integrated station/resource campaign with full-tail clearance and fault cases; one local controller and virtual detectors |
| Simulation and media | Retained 1 km six-car moving numerical run with repeated flexible spans and couplers, calculated-motion video, partial native detail CAD/images and visibility videos |
| Engineering acceptance | 0 accepted physical releases; design, manufacture and operational gates remain open |

Ring benchmarks and block-orientation seeds receive no railway lift or mass-saving credit. The moving run uses ideal independently allocated magnetic forces, rigid cars, linear beams and virtual sensing. Real force maps, shared-field phase, flexible bodies, lateral/soil/contact response, hardware timing and distributed controller handover remain open. The moving video uses calculated positions with declared amplification; other videos show final-position stock visibility, without collision or erection-strength acceptance.

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr design station`, `./osr design structure` and `./osr design protocol` reproduce their separate evidence. `./osr workbench` opens the review UI; `./osr book` builds the current reader edition. The scheduled runner refreshes implemented evaluators after source changes; it does not invent physical evidence.

Open the Workbench, review calculations, CAD and videos, read implementation and open work, inspect moving six-car evidence, inspect the manufactured dossier, inspect wider-stop corridor evidence, inspect booster and rescue quantities, compare Baghdad interchange alternatives, review v0.5.2 development notes, or download the reader book.

Published v0.5.1 remains preserved. This document describes v0.5.2 development on main and does not announce a new release.

# OpenSourceRail v0.5 fixed-zone ferrite-maglev engineering

engineering/ferrite_maglev/README.md

OpenSourceRail develops an open, locally buildable urban railway with fixed linear motors at stations and, where needed, located boosters. Trains coast between powered zones while onboard ferrite support, guidance and passenger services remain independently powered; passengers transfer between fixed lines at junctions. Simple vehicles, repeated civil works and repairable local production aim for high useful passenger throughput and affordable construction, operation and debt repayment. Current studies and digital workflows support development; an accepted physical railway has not been established.

Project introduction · Engineering approach and current status

The primary design uses fixed linear synchronous motors at stations for launch and receiving braking, plus located midway acceleration on longer legs. Guideway outside these finite powered zones remains unpowered. Onboard ferrite PM-biased active support/guidance and services storage remain independent. Structural GFRP sandwich bodies, repeated narrow concrete guidebeams and Rust control packages are the leading studies. Accepted engineering and physical releases: zero.

## Detailed candidate and remaining design gates

The coordinated design layer supplies explicit physical candidates, source-bound evaluations, bounded agent tasks, restartable numerical jobs and separate next-experiment/integrated selections. Selected beam sections and spans now propagate to manufacturing, civil material, BIM and construction quantities. Prestress uses local moving-load stress envelopes, and procurement availability constrains first material consumption. The comparison Workbench shows failures and missing installed equipment alongside calculated margins.

The new detailed-system packet uses one exact candidate for dimensioned magnetic cores/windings, mass and inertia, actual Rust lift/guidance currents, structural/civil loads, quantitative purchasing, native lifecycle and IFC/IDS/BCF field exchange. Current accounted mass is approximately25.6t tare/37.6t AW3 per car. Corrected25mm return-rail flux areas fail required AW3 lift within current/material limits. All three passive propulsion-band alternatives also retain failed applicable field/demagnetisation evidence. This blocks adoption and a new complete-service/finance claim. The old94ad actuator pass is retained as a separate configuration.

Actual native module and visibility video · Continued fault trajectories · Structural alternatives · Quantified manufacturing.

The owner/builder workspace restores city stop/service/OD edit proposals and16 named GIS layers. Its immutable compiler supplies generic planned workfronts, resources, procurement and monthly cash to the retained ERP adapter. Missing prices and independent physical acceptance remain explicit.

## Run and inspect

The connected Baghdad pilot joins controlled product accounting, coupled actuator experiments, launch segmentation, construction/procurement cash, access and service modelling, and serial-specific native business execution. Earlier mass targets are retained as benchmarks; the component-accounted product exposes its overrun and missing equipment definitions.

The migration register accounts for the actual v0.4 feature table. The first adoptable product is the owner/builder/operator planning and training platform, with separate reference hardware and GOA4 control products. The assurance profiles distinguish source trace, executed verification and independent acceptance.

```
./osr build
./osr design detailed-system
./osr design candidate-native
./osr design full-dynamics
./osr design operations-crosscheck
./osr plan serve
./osr design station
./osr design structure
./osr design protocol
./osr design corridor
./osr design booster-kit
./osr design line-capacity
./osr design stopped-recovery
./osr design drives
./osr design hub-access
./osr design qualification
./osr design assembled-body
./osr design body-stiffness
```

```
./osr design shared-hub-layout
./osr design pilot-product
./osr design pilot-actuators
./osr design launch-optimisation
./osr design delivery-schedule
./osr design pilot-corridor
./osr design pilot-variants
./osr design pilot-lifecycle
./osr design pilot-delivery
./osr design iterate
./osr design check
./osr design verify
./osr city baghdad
./osr sim baghdad --cycles 2
./osr workbench
./osr test
./osr book
cargo test --locked -p osr-maglev-control --all-targets
```

Current report · Workbench · city directory · architecture.

`./osr capabilities` publishes executable sources and scope. `./osr catalogue plan --city baghdad` resolves a selected regeneration, and `./osr catalogue run --city baghdad` rebuilds that city. `./osr readiness` reports source and acceptance gates. `./osr design release-check` remains held while engineering acceptance is open.

The dependency runner verifies implemented native, numerical, control, city and publication prerequisites. An interprocess lock excludes concurrent timer/manual builds. Progress is retained under `build/maglev-workflow/latest.json`. The runner rebuilds stale evidence and holds dependent work after failures; missing evaluators and physical measurements remain additional engineering work.

The stopped-controller recovery study executes fresh-epoch reconciliation for stationary consists at declared berths. It retains complete checkpoints and event histories, repeats every replay during verification, and requires new contracts after recovery. Moving recovery, multi-controller handover, authenticated installed storage and physical interfaces remain open.

The support/launch construction design defines the provisional coil and conductor quantities, local current/commutation software, converter/sensor purchasing families, winding schedule and electrical connections. `drive_design.py` derives R/L/material, phase uncertainty, fixed-group voltage/current, copper heat and DC/braking duties on current urban/suburban profiles. It executes `drives.rs` against an independent RL/alpha-beta coupon plant. Those original RL coupons remain separate from the old ideal-force moving train. The later pilot actuator and dimensioned-candidate dynamics couple actual Rust current loops to their own declared nonlinear plants; neither inherits physical magnetic validation. SBC supervision, local MCU loops and independent hardware protection/power remain separate; the larger winding's failed fit and magnetic/electronic qualification holds stay visible.

# Current inputs and design records

| Concern | Source |
|---|---|
| Architecture | Infrastructure policy |
| Vehicle/civil configuration | Configuration, reader, primary inputs |
| Wider-spacing operating corridors | Urban/suburban model, actual trajectories, Rust replay and limits |
| Booster and independent recovery hardware | Kit model, quantities and alternatives, native kit |
| Station motion and electrical plant | Operating model, published trajectory/overlap/power study |
| GFRP, magnet orientations and civil production | Structure model, study inputs, published quantities |
| Requirements and allocation | Requirements, systems register |
| Costs and manufacture | Process model, planning rates |
| Magnetic/native geometry | Geometry, field controls, native exporter, station-native packet |
| Station reservation protocol | Offline protocol evidence, persistent Rust interlocking ownership |
| Moving six-car twin | Numerical model, run/controls/video |
| Manufactured details | Dossier, specifications/margins/obligations, native solids |
| Integrated station/control campaign | Executable campaign, command/input/output evidence |
| City geography/service | Each city's `city.json`; compiler, operating-plan generator |
| Baghdad interchange alternatives | Investigation model, unselected access/shared-site comparisons, map |
| Interchange access engineering | Model, connector/support and passenger-flow alternatives |
| Sectioned multi-train prototype | Model, actual Rust reservations and adverse controls |
| Qualification tasks and physical records | Task/assessment engine, requirement chains, rig procedures and raw schemas |
| Assembled structural candidate | Generator and native checks, fitted regions, load-path changes and partial mass/inertia |
| Longitudinal stock alternatives | Repeatable generator and verifier, drilled stock, added full-AW3 mass and actual moving comparisons; no joined-body or default adoption |
| Shared-hub station geometry | Bounded fitter, directional approaches, platforms and actual door paths |
| Portable evidence bundles | Streaming multipart writer/verifier, command; exact committed sources, bounded archives, no physical-release authority |
| Onboard support control | Rust package, offline host profile |

Baghdad uses six 17 m bodies with 0.5 m assumed coupler gaps. The retained 30 kWh tare target is 14.2 t/car, including a 5.2 t equipped GFRP body, 3 t ferrite and an existing 1.8 t growth reserve. The 50 kWh operating reference is 14.4 t tare / 26.4 t AW3 per car (158.4 t six-car AW3); the pack increase is included once in mass. Four passenger doors per car are the body takeoff. The manufactured dossier uses that heavier case for hardpoints, passive supports, civil demands and exact brake energy/thermal quantities, alongside a named 30 kWh benchmark. Its counted primary-frame overrun and heavier brake proposal consume growth reserve; battery mass is not charged a second time. Unweighed hardware, joined-body behavior, fatigue, CG and inertia remain open.

The primary ordinary box uses 25 m spans, 0.80 m width, 100 mm webs, 150 mm flanges and end infills. Selected depth, stiffness and mass derive from current configuration and retained numerical evidence; the supplied 1.80 m case is a comparison. The detail dossier adds prestress/handling alternatives, one fixed and one guided bearing per span, longitudinal brake-load distribution and hollow-support/footing sensitivities. A larger footing is a conditional priced proposal under declared soil/load assumptions, not a geotechnical selection.

## Implemented checks and open boundaries

The qualification programme assigns each applicable requirement an exact revision, deliverables and prerequisites through physical measurement and independent acceptance. It provides specimen/rig procedures and typed raw-data assessment; prepared procedures and synthetic examples remain unaccepted. Use `tools/automation/osr-python -m engineering.ferrite_maglev.qualification --help` for the evidence assessment entrypoint. The multi-train and hub-access packets are additional studies; they do not select a hub, approve protection or change city defaults from an isolated passing scenario.

The separate Baghdad interchange investigation leaves the default273 stops and10 spacing conflicts unchanged. It compares24 extended-transfer placements on unchanged guideways and bounded disjoint network scenarios, retaining rejected extent controls and overlapping-pair constraints. Seven coordinated shared-site concepts instead draw changed guideway chords to a common location. None is selected or physically qualified. The original shared-site packet leaves tangents, levels, structures and internal walks undefined; its100m walk remains a sensitivity assumption. The separate shared-hub geometry study now fits candidate approaches, powered zones, platforms and actual door paths, retaining failed layouts and open structural/site acceptance. Extended-link quantities and electrical target credits exclude complete connectors, station civil and passenger-time economics. No equal-capacity savings or compact-hub proof follows from grouping IDs.

| Capability | Implemented scope | Evidence still required |
|---|---|---|
| Station trajectory | Jerk-limited launch, zero-thrust coast and receiving brake; energy closure and assumed magnetic-band overlap | Measured drag/magnetic losses, actual grades/wind, useful thrust, voltage/current/thermal limits and robust arrival margins |
| Station electrical costs | Fixed-group/section alternatives and separate power/charging target-rate rows | Configured equipment, supplier quotations, grid capacity, real duty and complete station scope |
| Body/civil quantities | GFRP panel mass cases, interface ownership, beam/end-infill quantities, support/foundation screens and target costs | Orthotropic structural analysis, member/joint/prestress/rebar detailing, soils, tolerances and construction-stage acceptance |
| Magnet orientations | Point-dipole ring reference and finite-block orientation inventories | Nonlinear finite-geometry steel/coil force/loss/demagnetization maps and measured shared-field behavior; no ring-gain credit |
| Native geometry | Partial panel/interface/civil/station definitions, export/reopening checks, retained images and visibility sequences | Complete production geometry, joints, equipment, drawings and qualified manufacturing tolerances |
| Rust support control | Module allocation, virtual replay, protection interfaces and support-preserving stop obligations | Actual magnetic/sensor plant, independent protection, physical I/O and measured timing |
| Launch ownership | Offline corridor/receiving-berth reservation, committed occupancy and full-clearance protocol checks | Authenticated complete-consist sensing, real receiving readiness, independent stopping and qualified system integration |
| Moving six-car replay | Six independent rigid-car states, actual Rust support allocation, repeated vertical flexible spans, longitudinal spring/damper chain, numerical controls and rejected comparisons | Measured magnetic maps, qualified shared-field phase/normal-thrust coupling, flexible bodies and physical timing; later candidate lateral/torsional/joint/contact studies have their own unqualified scope |
| Manufactured details | Counted frame stock, supported panels, joints, couplers, stator/bearing interfaces and handling/emergency load paths | Orthotropic laminates, joined load transfer, tolerances, fatigue/crash, calibrated limits and independent qualification |
| Integrated campaign | Actual Rust resource ownership and station control over incoming trip, dwell, next departure and negative cases | Authenticated sensors/plant, distributed handover, real receiving readiness and independent physical protection |
| Adaptive design | Previous-result-guided vehicle/beam/gain proposals, ancestry, Pareto observations and separate confirmation | Completed parts and qualified evaluators/plant before manufacturing or operating selection |
| Factory and city delivery | Conditional two-shift bottleneck capacities, duty/fleet, route quantities and process needs | Measured cure/crew/logistics rates, surveys, route recovery, complete economics and accepted instances |

The adaptive search, configured in iteration.toml, has explicit evaluation/stagnation budgets and exact-source restart receipts. Failed confirmation disqualifies the exact revision. Published history retains proposals, ancestry, traces and recommendation. Mass/inertia allocations and preference weights remain research assumptions. An analytical recommendation does not automatically change acceptance criteria or reference configuration.

# Operation, manufacture and costs

The primary operating strategy targets urban stops at least 1.5 km apart and about 2 km on average, with 4–5 km suburban legs and located midway boosters where required. Fixed-line junctions are passenger transfers, without physical train-routing switches. The retained 1 km/88s/180s component benchmark is distinct from those longer operational profiles. The source-bound station-powered packet preserves the supplied 90 s/190 s benchmark. Actual city intervals and headways derive from bounded motion, overlap and full-tail-clearance searches. Stops are placed at passenger transfers and appropriate intervals, with retained access assignments. Nearby mandatory transfers that violate minimum spacing are reported explicitly; straight-chord alignment candidates preserve crossing nodes but do not approve clearances, bends or land. Terminal-loop proposals remain separately unqualified. Original access counts, operating stops, selected capacity and any shortfall are explicit.

A committed train in a coast or booster corridor keeps protected resources until positive full-consist clearance; timeout cannot revoke physical occupancy. The actual Rust campaign joins this ownership to station control for two consecutive legs, and retains receiver/sensor/drive/resistor/stale/phase/uncommitted fault cases. Virtual detectors and one local controller do not establish distributed handover. Independent friction-brake and winch recovery proposals now have force, energy, reserve and civil load-path calculations; calibrated friction/thermal hardware, access, curved recovery and physical stopping/holding remain unqualified.

At 60 kW/car hot services, the authored 90 s trip plus 15 s disconnected dwell consumes 10.5 kWh/train. Replacing this in 45 s takes 840 kW plus 360 kW direct services, or 1.20 MW/train before losses. The retained benchmark uses240 kW/car input and90% assumed conversion. Longer operating cases evaluate their own charger, battery, dwell and repeated services-energy requirements; those ratings are not inherited from the1km benchmark. The failing 200 kW input comparison is retained. Charger rating, taper, ageing, thermal duty and grid capability remain open.

City priced-scope subtotals separately label fleet, ordinary elevated guideway, terminal-loop civil work, station electrical targets, a conditional larger-footing increment, located booster plant and independent rescue kits. The footing delta accounts for concrete, reinforcement and earthworks under an unqualified soil/load proposal. The authored USD 4.255 million versus USD 3.313 million ordinary route-km comparison remains a narrow historical target-rate comparison, not the price of the new complete operating plan. No supplier quote or complete installed programme price exists. Stations, land, depot, external grid, evacuation, factories, acceptance, recovery, finance and other unpriced scope remain additional.

The booster kit counts2.5 m face cassettes,12.5 m electrical sections, mounts, bolts, cabinet pads and actual spatial reactions at fixed/guided bearings. Equivalent $I^2R$ duty is integrated without inventing winding temperatures. Complete winding/core/cooling masses and nonlinear magnetic/electrical/thermal maps remain open. The partial native scene contains defined mount/drum/portal/anchor stock, without claiming a complete motor or assembled interference check.

Recovery alternatives compare ≤1km accessible node spacing with midpoint-only nodes,30/50/60 kWh packs and0.5/1 m/s towing. The5 km/300 s hot study retains the failing30 kWh midpoint-only slow-winch case. Standalone rope/drum/portal/foundation, rated drive/power allowances and QA/access costs are additional; co-location with an existing station, pier or booster earns no automatic saving. Cable deployment, certified terminations, essential power, actual soils, rescue/evacuation access and assisted passenger operation remain physical gates.

City manufacturing records carry material quantities, process routes, equipment needs and construction approaches. Candidate processes include controlled GFRP infusion/RTM and bonding, precast mould/cage/prestress/cure work, reaction-iron and carrier fabrication, station coil winding, cartridge assembly, wiring and tests. The production model takes the minimum of factory, foundation, pier, delivery, erection and final survey capacities. Its 56.25 route-m/planned-day target is not measured output. Demould, connection strength, crane-at-radius, ground and final QA gates remain unexecuted.

Prefer verified British manufacture when considering UK machinery. A UK supplier or integrator does not establish manufacturing origin. Complete local content remains unknown until origin-traced bills and configured offers exist.

# Evacuation and publication

The preferred research layout has one beam per direction without a continuous external walkway. Necessary magnetic faces, supports and connections remain. Vertical spiral chute research covers variable height, anchorage, landings, assistance, fire/weather, partial failure, rescue and driverless use. Walkway removal needs accepted route and onboard alternatives at every allowed stopping point.

The moving-train video uses calculated six-car positions and beam states with declared visual amplification. Legacy CAD videos show final-position visibility; the assembled-body movie adds illustrative installation translations. None establishes collision-free erection, crane performance or chute deployment. CAD integrity, PDE residuals, virtual controller ticks and consistent publication hashes each establish only their checked digital scope. Closure work and v0.5.2 development notes remain explicit.

# UK power-module procurement

Checked 22 October 2026. This supplements the 40-line construction BOM; it changes procurement preference, not quantities or the executable electrical assumptions. No order line, quotation or converter is selected.

| BOM family | British manufacturing candidate | Imported comparison | Confirmation needed |
|---|---|---|---|
| LSM-IGBT: topology-specific switch modules; old96-module count is not current procurement | Dynex 3.3 kV IGBT family | Infineon FZ1500R33HE3BPSA1, 3.3 kV/1500 A catalogue reference | Exact part, manufacturing site, lifecycle, switching/thermal duty, SOA and parallel sharing |
| LSM-BRIDGE:140 groups/station in the retained5m-group/350m-per-direction actuator case; new1350A group remains unqualified | Dynex custom power-assembly capability; UK manufacturing scope to specify | Nidec M700 remains an industrial integration reference, without a selected 2400 V catalogue configuration | Complete topology, UK operations, gate/protection design, cooling, insulation, testing and quote |

Dynex's current company page describes in-house UK manufacturing and semiconductor fabrication/assembly capability. It also states that higher volumes can be outsourced to China. Treat Lincoln manufacture as a procurement requirement for the proposed order, not an automatic property of every Dynex-branded device. Its current finder includes 3.3 kV modules, but an exact suitable part remains unselected. Dynex company and manufacturing, current module finder.

The imported comparator is a single-switch 3300 V/1500 A module, with a 190 × 140 mm package. Infineon's product page lists Hungary for assembly and Malaysia for diffusion for this ordering code; confirm the delivered lot's records. These catalogue values do not establish an OpenSourceRail current, thermal or switching rating. Infineon product and origin fields.

Four parallel devices per switch position is a provisional count. Do not reduce it merely because a catalogue current exceeds a divided peak demand. Check current sharing, switching energy at the actual voltage/current/frequency, diode duty, junction/case temperatures, cooling and fault clearing. Receiving-chopper modules have their own duty and are not silently included in the traction-bridge count.

The proposed bus remains 2400 V DC. A lone 1700 V device remains unsuitable as its two-level switch; a 3.3 kV label alone does not qualify transient margin or insulation. The construction study retains those gates and its current/voltage/heat comparisons.

Exact UK-origin evidence, imported input value, local assembly and Iraqi local content remain separate records. No localisation percentage or financing eligibility is inferred from a supplier's address or ownership. Engineering, manufacturing and physical acceptance remain open.

Browser execution records can be refreshed after the three development profiles and `./osr build` pass:

```
tools/automation/osr-python tools/automation/retain-pilot-ui.py prepare
node tools/automation/tests/verify-maglev-workbench.mjs build/maglev-workbench-browser > build/pilot-final-browser-
workbench.log 2>&1
node tools/automation/tests/verify-maglev-v05.mjs build/maglev-browser > build/pilot-final-browser-reference.log 2>&1
tools/automation/osr-python tools/automation/retain-pilot-ui.py retain
tools/automation/osr-python tools/automation/retain-pilot-ui.py verify
```

The retainer checks unchanged publication and UI inputs before retaining screenshots, logs and checksums under `engineering/assurance/v052-pilot-ui/`. These are digital UI checks, not engineering acceptance.

# osr-maglev-control

crates/osr-maglev-control/README.md

OpenSourceRail develops an open, locally buildable urban railway with fixed linear motors at stations and, where needed, located boosters. Trains coast between powered zones while onboard ferrite support, guidance and passenger services remain independently powered; passengers transfer between fixed lines at junctions. Simple vehicles, repeated civil works and repairable local production aim for high useful passenger throughput and affordable construction, operation and debt repayment. Current studies and digital workflows support development; an accepted physical railway has not been established.

Project introduction · Engineering approach and current status

Portable fixed-zone ferrite-maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Station and located-booster LSM demand is separate from onboard active PM support and guidance. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

`drives` adds a separate local-loop software layer: bipolar support-current PI, three-phase Clarke/Park transforms and bounded dq current regulation for station LSM converters. It checks dual current acquisition, phase uncertainty, deadlines, temperature, DC voltage, protected authority and receiver/resistor readiness. Faults latch; reset requires fresh healthy zero-demand stopped/secured evidence. An unavailable support command requests external local fallback; a wayside fault never requests onboard support shutdown. None of these outputs operates hardware.

The proposed division is SBC supervision/allocation, local MCU current and gap loops, and independent hardware protection plus support power. The 20 kHz target is an input, not a measured deadline. `cargo test --locked -p osr-maglev-control --test drives` runs nine regression tests. `cargo run --locked -p osr-maglev-control --example winding_replay` emits three actual 4000-frame RL/FOC coupon replays and fourteen adverse controls. The independent plant uses a standard-library trigonometric oracle. The 40 A coupon does not qualify a multi-kA launch converter or close the ideal-force moving-train model.

See the construction design and counted BOM for support-coil packages, winding/phase schedules, power-electronics families, sensing, connections, supplier references and the remaining magnetic/hardware acceptance gates. No PCB, ADC/PWM adapter, gate-drive or installed timing is released.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

## 24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 support stations. This is one rigid car; the complete moving six-car Baghdad train and a validated six-degree-of-freedom suspension remain open. The array model has no longitudinal actuator; nonzero longitudinal demand appears as an unachieved residual. The separate station module supplies an offline LSM demand screen, without closing physical propulsion/braking, traction/support coupling, flexible vehicle/beam modes, finite-angle dynamics or contact behavior.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis

success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

# Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is a retained numerical comparison: 18,750 kg tare plus 120 × 75 kg passengers. The current station-powered design instead targets 14,200 kg tare; this retained control case is not the new vehicle mass closure. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

# Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes the current configured section and native quantity-derived line mass, with 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame,

and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

## Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

## Station launch and receiving study

`station` evaluates bounded offline launch, coast, receive and secured modes. It checks exact configuration/zone identity, fresh monotonic frames, phase uncertainty, reservation assertions, magnetic-band overlap and assumed current, voltage and thermal limits. Coast demands zero station thrust. Receiving braking requires the declared brake resistor to be ready. Receiver loss after commitment requires independent stopping/recovery; clearing a Boolean drive demand does not stop an already coasting train.

```
cargo run --locked -p osr-maglev-control --example station_replay
./osr design protocol
```

The scalar force/current relation, winding R/L/flux and voltage/loss calculations are explicit hypotheses. They are not measured motor maps, three-phase hardware commutation, shared support/thrust qualification or a six-car electromagnetic simulation. An external `launch_committed` assertion is not an authenticated interlocking or detector record. Persistent corridor/berth ownership is exercised separately in `osr-interlocking`; permission expiry cannot release a committed train, and full-consist clearance is required. No physical output adapter or railway operating authority is supplied.

The longer-corridor study (`./osr design corridor`) separately evaluates urban 2 km and suburban 5 km profiles, finite station/booster overlap and electrical rating alternatives. A booster is a nonpassenger powered asset; junctions are passenger transfers between independent fixed lines. The 1 km station replay remains a component benchmark. Its passing result cannot clear longer-profile voltage, service-energy, recovery, section-handover or exclusive-reservation capacity holds.

## Adaptive research tuning

`./osr design iterate` varies bounded proportional and damping gains alongside vehicle mass allocations and guidebeam sections/spans. The bridge accepts mass and gain scales as optional command arguments; nonfinite or out-of-bound scales are rejected before processing frames. Exact revisions face separate confirmation scenarios and replay/excess-gap negative controls. This offline tuning establishes neither measured magnetic response nor stability over the full operating envelope.

# OpenSourceRail architecture

docs/ARCHITECTURE.md

OpenSourceRail v0.5.2 development uses fixed-zone ferrite maglev. Fixed station linear synchronous motors launch and receive each train; located midpoint accelerators are alternatives for longer suburban legs. Coasting sections outside these finite powered zones have passive magnetic reaction iron. Onboard ferrite PM-biased active support and guidance remain powered independently, alongside passenger services, protection and emergency equipment.

The objective is broad city access with compact elevated guidebeams, fewer moving running-gear parts and practical local manufacture. A 99% elevated-mainline planning target still depends on route rights, utilities, ground, curvature, station access, evacuation and recovery.

## Current detailed candidate

The candidate chain accounts approximately25.6t tare/37.6t AW3 per car and carries one revision through magnetic geometry, currents, mechanical/civil loads, manufacturing, native purchasing and BIM/field records. Corrected thin return-rail saturation prevents required AW3 support; propulsion demagnetisation/convergence and structural/protection failures also remain. No complete train, manufacturing release or cost advantage has been established. Earlier base and94ad actuator configurations remain identified comparisons, not interchangeable evidence.

The planning workspace supports city proposals independently of hardware acceptance; its generic delivery adapter and authenticated business navigation do not grant construction or motion authority.

## Configuration and interfaces

Baghdad uses six 17 m bodies. The 104.5 m coupled footprint is an assumption until couplers and gangways are resolved. The historical target study targets 14.2 t tare/car, including a 5.2 t equipped GFRP sandwich body, 3 t ferrite, 30 kWh services storage and 1.8 t growth reserve. Four doors per car are the body-study basis. No complete weighed body, load-bearing joint set or qualified magnetic circuit establishes these mass targets.

The ordinary civil candidate is a 25 m, 0.80 × 2.50 m box with 100 mm webs, 150 mm flanges and end infills. Hollow columns and caps are explicit quantity studies. Shallow, single-shaft and pile-group foundations remain alternative families, selected only with ground and structural evidence. Reaction iron, anchors, bearings, joints, rail survey and all construction stages remain in scope.

| Layer | Controlled definition | Implemented evidence and boundary |
|---|---|---|
| Service and city | Geography, demand, formation, runtime, headway and station spacing | Urban minimum target 1.5 km and mean target 2 km; explicitly classified suburban legs 4–5 km with booster alternatives; short interchange conflicts remain visible |
| Mechanical | Structural sandwich panels, four doors, ferrite cartridges, support/guidance and services | Fitted structural/material regions with native interference, drilled hardpoints and partial mass/CG/inertia; complete equipment, joined behavior and qualification open |
| Fixed-zone propulsion | Station/booster LSM cassettes/groups, phase sensing, bridges, DC supply, receiving braking and charging | Separate longer-leg jerk/overlap/rating/service/protection comparisons; measured force/current/voltage/thermal maps unqualified |
| Civil | Guidebeams, supports, foundations, stations and interfaces | Exact section/end-infill quantities, gross elastic screens and foundation sensitivities; no accepted strength or soil capacity |
| Software | Independent support control and persistent launch/corridor/berth ownership | Offline Rust tests/replays; no qualified sensing, physical actuator adapter or measured hardware timing |
| Industrialisation | Moulds, cure/release, crews, transport, cranes, joints and survey | Conditional bottleneck capacity; no measured production or supplier-price evidence |
| Acceptance | Exact configuration, source, unit/serial identity, tests and reviewers | Requirement task chains, typed physical test protocols and source-bound receipts; engineering/manufacturing/physical acceptance remains open |

Infrastructure policy, configuration, operating study and structure study control the interfaces. City `city.json` inputs feed the same current compiler. Junctions are passenger transfers between independent fixed lines, with no mainline train-routing switch. Straight alignments are the design objective; retained mapped routes and unselected straight-chord alternatives must keep their site and cost evidence separate.

# Launch, receiving and recovery

The electrical construction package separates three control responsibilities. The car SBC runs supervision and force allocation; proposed local MCU boards acquire gap/current signals and run fast loops; independent protection, essential support power and capture handle faults. Permanent magnets provide bias flux, while active windings correct unstable attractive suspension and guidance. An ordinary SBC process is not a qualified 20 kHz power-stage controller. The new Rust `drives` layer executes bipolar current PI and sensored three-phase dq commutation in offline coupon plants. It has no ADC/PWM adapter or physical-output authority.

The counted construction candidate has one lift coil and two series-opposed guidance coils per cartridge: 432 coils and 288 bipolar circuits per six-car train. Its 50-turn compact coil passes declared depth/packing and current screens, while retaining rejected larger-pack and lower-turn alternatives. Separate wayside LSM bridges use indexed three-phase winding groups. Their BOM, winding schedule and connections retain material, converter, sensing, DC and braking quantities. Measured magnetic maps, complete wiring/PCB drawings, thermal design, installed timing and physical actuator validation and complete routing/thermal design remain open. Later candidate dynamics integrates actual currents with its declared geometry-derived plant; that does not qualify this older50-turn construction case. A station propulsion fault must not remove onboard support.

The UK procurement supplement adds Dynex's British manufacturing capability as a first candidate for launch semiconductors and custom assemblies, retaining an imported comparator and requiring exact order-origin confirmation. It does not select or qualify a part.

A launch reserves the interstation corridor, receiving berth and protective boundary for one train. Permission expiry does not release committed occupancy. Release requires complete-consist clearance evidence; lost communications or a timer cannot establish that clearance. The receiver must remain able to stop a train after departure. Independent stopping/holding, receiver faults, low-service energy and stranded-train recovery require their own engineered designs.

The separate `osr-progressive-departure/1` candidate instead protects a finite stopping envelope before a stopped departure and renews it from fresh evidence. Its local multi-train replay includes withheld arrival permission and explicit reaction/braking transients. It does not replace `StationLaunch` or the city capacity basis. Restored candidate snapshots remain fenced; operational restart, distributed handover and physical brake/detector qualification remain open.

Coasting feasibility depends on grade, drag, wind, launch speed and arrival margin. An unpowered span cannot restart a stationary train from a distant station. Terminal return geometry, depot movement and rescue equipment remain explicit requirements. The nominal trajectory does not close these obligations.

The 1 km/90 s authored and 1 km/88 s developed runs remain explicit benchmarks. They do not verify the new 2 km/5 km operating design. Longer profiles compare converter ratings, charger power, dwell and larger services packs with their mass and cost changes. Whole-leg exclusive reservations impose a capacity constraint; a passing motion calculation cannot clear it. Station and booster power/charging are separately priced targets from the incomplete civil subtotal. Complete network economics must include both, plus stations, fleet, recovery, land, qualification, factory capital and financing.

# Evidence boundary

Ring-field controls and block orientation seeds do not demonstrate railway lift, thrust or magnet-mass gains. Gross beam sag is distinct from relative magnetic-gap error. The stationary single-car demonstrator is retained separately from the moving six-car Rust/beam/coupler model. Both use ideal force channels. Longer operational profiles need their own moving confirmation; flexible bodies, torsion, joints, soil and calibrated electromagnetic/thermal response remain open.

Native CAD covers partial assemblies. Visibility videos show final-position parts and an illustrative sequence; they do not establish collision-free motion, erection or chute deployment. Continuous-walkway removal remains conditional on accepted onboard and route evacuation, including assisted access and safe landings at every permitted stop. All release flags remain false. Accepted engineering and physical releases remain zero.

# Roadmap

docs/ROADMAP.md

OpenSourceRail v0.5.2 development coordinates fixed station/booster-zone ferrite maglev across mechanical, civil, software, manufacturing and city planning. Station LSMs launch and receive the train; longer suburban legs compare located midpoint accelerators. Independent onboard support and passenger services remain active during the unpowered coast. Junction stations transfer passengers between fixed lines. Urban spacing targets 1.5 km minimum and approximately 2 km average, with explicitly classified suburban 4–5 km alternatives. Implemented work and the current report identify the tested digital scope. Accepted engineering and physical releases remain zero.

## Primary maglev closure queue

The coordinated design loop implements the first carrier/guidebeam campaign. Its development packages are tracked below; implemented software does not close physical requirements.

| Package | Implemented in this iteration | Remaining completion gate |
|---|---|---|
| Dependable evaluators | Local moving-load prestress envelopes; procurement before first consumption; explicit beam/span selection; regression cases | Material, joint, supplier and site correlation |
| Common contracts | Physical, evaluation and selection identities; complete/unknown inventory; source-bound registry; rigid transforms and conservative stock sweeps | More physical families and complete installed inventories |
| Domain adapters | Current nonlinear magnetic plant and actual native structural/civil chain | Independent spatial field benchmarks, installed joint models, repeated electrothermal duty |
| Campaign runner | Bounded tasks/dependencies, numerical workers, resource limits, restart, immutable evidence and failure-directed revisions | Broader capabilities and measured input ingestion |
| Multi-option comparison | Three carrier/beam families, bounded revisions, Pareto next experiment and separately gated integrated ranking | A candidate surviving simultaneous support, guidance, thermal, mass and operation requirements |
| Research twin | Selected native CAD/STEP, BOM, IFC/IDS/BCF, assembly scope and material-constrained draft delivery | Complete installation, manufacturing release and full route/fault confirmation |

The primary context remains six cars and 180-second headway for 14,400 normal passengers/hour/direction. Changed speed, station spacing and formation are explicit research alternatives, with complete service and civil consequences. The immediate experiment priority remains geometry-resolved magnetic forces on the same physical stock, followed by counted joints/cooling and coupled gap, temperature and deformation maps. No integrated survivor is implied by a completed digital campaign.

The design completion execution plan preserves all 576 qualification tasks across eight workstreams and orders the next magnetic, propulsion and curve-layout experiments. Additional bounded optimisation has executed 240 magnetic cases with no joint-screen survivor and 28 propulsion duty comparisons with 17 conditional coupon passes. Neither changes the canonical candidate or completes the four-stop electromechanical journey. Exact-source audits and negative controls accompany the new results.

The next executed design round adds 256 counted support/cooling cases, 18 separate-guidance material bounds and 160 finite-carrier/curve cases. Service comparisons vary speed, station spacing, dwell and formation at both the current comparison mass and the proposed support material-mass floor. Normal passenger capacity, complete tail clearance, charging, added platform/powered-zone scope and thermal failures stay visible. The research specification records these alternatives; no integrated configuration or low-cost advantage is accepted. README coverage maintains current introductions without altering frozen scientific packets.

The city briefs show current line/service tables, energy, local production routes and low/reference/high CAPEX/OPEX/debt sensitivities. The connected Baghdad pilot develops walking catchments, explicitly synthetic OD/accessibility, resource-constrained construction, procurement/monthly cash and serial asset/QA/maintenance interfaces for a selected current line. The native lifecycle campaign uses simulated assets in isolated business applications; it does not commission a railway. Extending that pilot across the whole catalogue, restoring the interactive GIS editor and independent SUMO comparison, and qualifying observed demand, sites and equipment remain separate work. Station-led GOA4 dispatch and the progressive-departure candidate retain their independent protection and acceptance gates. Whole-life affordability, owner-held design knowledge and domestic capability remain central selection criteria.

| Package | Evidence required for closure |
|---|---|
| Requirements and configuration | Approved six-car loads, service/grade envelope, door layout, tolerances and quantitative acceptance criteria |
| GFRP body | Sized orthotropic panels, longitudinal members, magnetic hardpoints, joints, openings and coupler/crash paths; fire, heat, fatigue, repair and weighed equipment closure |
| Magnetic assemblies | Complete shared or separately priced support/guidance/thrust circuits, cartridge retention, capture, sensors and tolerances |

| Package | Evidence required for closure |
|---|---|
| Magnetic force and loss | Converged finite-geometry/steel/coil maps over gap, phase, current and temperature, followed by rig correlation; no automatic ring-paper gain credit |
| Station and booster propulsion | Located fixed-group/section topology, phase registration, winding R/L/flux, current/voltage/thermal limits, receiving brakes, fault response, utilities and actual purchasing quantities |
| Service energy and charging | Verified hot/normal duty, battery life/reserve, complete dwell charging, power-loss support and recovery mission |
| Civil kit | Prestress/rebar/cover/end blocks, bearings, rail anchors, joints, hollow support connections, ground-specific foundations and every handling/construction stage |
| Launch and protection | Atomic corridor/receiving-berth reservation, persistent committed ownership, proved tail clearance, independent receiving readiness and stopping/holding under faults |
| Coupled dynamics | Moving six-car loads, flexible body, lateral/torsional response, curves, joints and soil with calibrated magnetic/electrical/thermal dynamics |
| Hardware | Isolated I/O, current drives, protection/watchdogs and measured end-to-end timing on selected SBC/MCU equipment |
| City delivery | Grade/coasting arrival margins, surveyed routes/utilities, terminal return geometry, depot/recovery access and evacuation at every stop location |
| Factory and erection | Measured mould/cure cycles, strength releases, delivery/crane-at-radius plans, joint completion, survey acceptance and bottleneck output |
| Economics and origin | Complete origin-traced MBOM, configured supplier offers, station/civil/fleet scope, capital/tooling, maintenance, financing and uncertainty |
| Demonstrator | Manufactured serials and exact revisions, independent inspections, measured system operation and physical acceptance |

The qualification programme maps the applicable requirements to dependency-ordered tasks, physical test procedures and guarded raw-evidence assessment. It supplies the execution route for this queue; every physical acceptance gate remains open until actual evidence and independent dispositions exist. New fitted structural CAD, multi-train controls and shared-hub geometry close defined digital studies while retaining their unresolved interfaces.

The implemented station study adds jerk-limited launch/coast/arrival calculations, magnetic-band overlap, receiving-brake duty and headway/power/cost sensitivities. The structure study adds quantity-derived GFRP cases, orientation seeds, five beam families, support/foundation screens and a two-shift bottleneck model. These are calculated candidates and production targets; they do not supply the missing structural, magnetic or measured factory evidence.

The adaptive designer uses preceding results to propose vehicle-mass, beam/span and actual Rust-gain variants. It preserves ancestry/traces and uses separate confirmation cases. Its original search plant is a stationary single rigid car; the moving six-car plant and longer-corridor confirmations are separate evidence. All currently use ideal magnetic force channels. A preferred candidate remains a bounded research result, without automatic promotion to accepted geometry or city criteria.

The dependency runner rebuilds stale implemented studies and holds downstream outputs after failures. Completing every automated check does not complete the engineering queue. Published v0.5.1 remains a distinct preserved release; this revision is documented in v0.5.2 development notes.

# Coordinated candidate design

engineering/design_search/README.md

OpenSourceRail aims for simple fixed lines, affordable construction and repair, local production and high passenger throughput. Ferrite support and guidance remain independently powered while fixed wayside motors propel the train. Passengers transfer between fixed lines. This package connects physical design alternatives and deterministic engineering calculations without granting physical acceptance.

One versioned candidate describes architecture, native stock, frames, materials, interfaces and installed equipment. Its physical SHA256 excludes modelling hypotheses and lineage. Each evaluation has a separate identity covering the requirements, sources, model hypothesis, settings, mesh, scenario and observed runtime. A selection has a third identity covering its purpose and declared policy. Results also carry a complete payload digest, so edited metrics cannot reuse an evaluation identity.

## Run the first campaign

```
./osr design campaign create campaigns/carrier-beam.toml
./osr design campaign run <campaign-id> --max-jobs 2
./osr design campaign run <campaign-id>
./osr design campaign compare <campaign-id>
./osr design campaign tasks <campaign-id>
./osr design campaign publish <campaign-id>
./osr twin compile <candidate-id> --campaign <campaign-id> --purpose research --allow-incomplete
```

The first run deliberately pauses after two completed jobs; the second resumes matching work. Omit `--max-jobs` for an uninterrupted run. State defaults to `build/design-search`, separate from source checkouts. A source revision creates a new campaign identity. Exact request matches can reuse cached results; tool errors and interrupted attempts remain retryable evidence. Job limits and the total campaign runtime budget are separate from agent conversation limits.

The declared campaign compares rigid existing hardpoints, four yaw carriers and five yaw/slide carriers with different guidebeam sections and 20/25 m spans. Failure-directed revisions remain within the declared stock/span bounds. It retains the six-car Baghdad context, 120 normal passengers per car and a 14,400 passengers/hour/direction comparison floor: 180-second headway. Larger formations or changed speed/spacing require a separate service comparison with platform, charging, fleet, evacuation and civil consequences.

## What runs

| Module | Responsibility |
|---|---|
| `contracts.py`, `registry.py` | Physical/context records, typed results, append-only identities and exact cache |
| `families.py` | Counted carrier stock, selected beam/span and bounded revisions |
| `evaluators/` | Definition checks, actual structural/magnetic calculations and explicit coupling hold |
| `geometry.py` | Proper rigid transforms, oriented stock checks and conservative sampled sweeps |
| `coupling.py` | Budgeted fixed-point state/residual iteration with explicit validity failures |
| `campaign.py`, `worker.py` | Dependency dispatch, private runs, resource limits, restart and provenance |
| `verification.py` | Independent stock arithmetic and source checks; no domain-physics confirmation |
| `selection.py` | Separate Pareto rankings for next experiments and integrated baselines |
| `twin.py` | Candidate-bound actual native CAD/STEP, BOM, assembly, IFC and draft delivery records |
| `publication.py` | Portable comparison Workbench, complete retained evidence and integrity check |

The current adapters call the native stock/structure chain and nonlinear magnetic plant. They do not select using the earlier ideal-force adaptive evaluator. Cheap definition failures remain visible; independent domain screens can still run to diagnose which architecture needs revision. Execution, numerical validity, engineering screen, evidence maturity, selection purpose and physical acceptance are distinct fields. A completed job may correctly report a failed screen.

Unknown bearings, slides, protection and equipment masses never become zero. Known stock gives a subtotal; complete mass, CG and inertia remain null when the installation is incomplete or unlocated. Strict twin compilation refuses these gaps by default. Explicit `--allow-incomplete` compiles a research package with those gaps listed. Native CAD contains one module coupon, one selected beam/span and per-car counted carrier stock in review poses. It does not depict a complete installed train. Assembly instructions identify unresolved retention, handling and removal details rather than claim a proven collision-free installation.

The current coupled adapter retains a hold because calibrated temperature, cooling, deformation and magnetic-gap maps are missing. The generic coupling solver can converge a supplied component model, but that does not establish a converged physical train. Arithmetic/source confirmation cannot satisfy the integrated policy's independent domain requirement. A no-survivor result is a valid completed campaign.

## Agents and numerical workers

Project instructions and eight role files in `.codex/agents/` assign systems, magnetics, electrical, mechanical, civil, twin, service and verification work. Agent tasks carry candidate/requirement identities, allowed variables, interfaces, evaluators, artifacts and budgets. Ordinary numerical workers execute parameter samples; agents develop and interpret evaluators. Source changes use isolated worktrees. Only the coordinator publishes.

Role instructions inherit parent permissions. Where available, the campaign runs its verifier using a read-only filesystem and isolated network through Bubblewrap, with its private output and temporary scratch writable. The attempt receipt records actual isolation. Source and result checks still run if isolation is unavailable; the receipt does not mislabel that fallback as an enforced sandbox.

## Verification and remaining engineering

```
tools/automation/osr-python -m pytest -q engineering/design_search/tests
tools/automation/osr-python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_candidate_structures.py
```

Meaningful regressions cover local prestress stress envelopes, lead times before first consumption, selected-span quantities, rigid transforms, unknown stock, identity mixing, result tampering, resume, worker errors and selection policy. Native integration tests reopen actual FCStd/STEP and parse actual IFC/IDS/BCF.

Next engineering work is independently correlated spatial fields on identical geometry, complete joint/cooling/protection stock and repair paths, measured material properties, current/thermal maps, coupled deformation/gap closure and then full formation/route/fault confirmation. No supplier price, material test, ground investigation or acceptance is invented by this software. The held qualification programme continues to control physical dispositions.

# Coordinated design research

docs/design-engine/design-search/README.md

OpenSourceRail develops simple fixed-line ferrite maglev for affordable construction, local repair and high passenger throughput. This campaign connects actual carrier stock, selected beam/span, current magnetic calculations and delivery quantities through one physical candidate definition.

Campaign `f60de0526feabfb9b2b9b590f9aaa53fe64d9c40cc22a6afe7d7f7f8f5853d9a` evaluated 5 physical alternatives in 25 primary calculations, retaining 5 independent arithmetic/source checks. The next-experiment selection is `ed05b47c9216df1819f745bef9597812a9d1ef009501cc399ae1da78904b5798`. Integrated baseline: no-eligible-candidate. Physical acceptance remains open.

Interactive comparison · Complete immutable evidence · Candidate contracts and reproduction

The research twin contains one module coupon, the selected guidebeam span and counted per-car carrier stock. Installed joints, cooling, protection, complete mass distribution, correlated field maps and coupled thermal/deformation behavior remain incomplete. Arithmetic/source confirmation does not independently validate domain physics. No city defaults or qualification dispositions are changed.

The six-car Baghdad context retains 120 normal passengers per car and a 14,400 passengers/hour/direction comparison floor, requiring 180-second headway. Formation and operating changes must carry their platform, charging, fleet, evacuation and civil consequences.

```
./osr design campaign create campaigns/carrier-beam.toml
./osr design campaign run f60de0526feabfb9b2b9b590f9aaa53fe64d9c40cc22a6afe7d7f7f8f5853d9a
./osr design campaign compare f60de0526feabfb9b2b9b590f9aaa53fe64d9c40cc22a6afe7d7f7f8f5853d9a
./osr design campaign publish f60de0526feabfb9b2b9b590f9aaa53fe64d9c40cc22a6afe7d7f7f8f5853d9a
```

This publication binds evaluator sources and campaign membership. A source revision requires a new campaign; interrupted jobs resume only with matching inputs. The digital jobs complete independently of the held physical qualification programme.

# First adoptable product: the delivery platform

docs/first-adoptable-product.md

OpenSourceRail has three linked products: the owner/builder/operator delivery platform, reference hardware and factories, and the open GOA4 control platform. The delivery platform can support controlled planning and training before a train or guideway receives physical approval.

The current adoption example is the Baghdad digital pilot. It connects current product revisions to procurement planning, simulated receiving and inspection, accountable rework, serial installation, equipment condition, maintenance, replacement and independent simulated handback. Its native lifecycle exercise uses an isolated ERPNext/FUXA/Workbench deployment. Existing owner/operator sites retain their own data and permissions.

A builder should be able to follow a physical unit from its product revision and drawing/BOM references to its order, inspection, installed position and replacement history. A failed or stale inspection must remain a hold. Business software must not accept its own hardware, waive independent review or authorise train movement.

The capability migration register accounts for every feature in the v0.4 feature table. It distinguishes current tested scope, replaced implementations, disconnected functions and missing capabilities. The separate editable planning workspace now supplies16 named GIS layers, immutable stop/service/OD proposals and a generic planned delivery compiler. The independent SUMO comparison executes railway timetable/block/dwell controls on the same stops; it does not model magnetic actuators. Whole-city BIM, physical field acceptance and full OCC/HR federation remain incomplete.

For a real deployment, the owner supplies legal companies, controlled calendars, business masters, user roles, approved procurement and QA procedures, equipment bindings, data retention and recovery responsibilities. Physical inspections require authenticated measured evidence and independent acceptance. The repository's simulated serial records establish development behavior only.

# Baghdad connected digital pilot delivery

docs/design-engine/ferrite-maglev/pilot-delivery/README.md

Connected provisional digital pilot. Packet integrity, implemented design resolution, physical verification and independent acceptance are separate states.

Verified digital packets: 9. All three development profiles executed against current sources: False. Physical releases: 0.

| Deliverable | Digital state |
|---|---|
| migration-register | delivered |
| controlled-configuration | delivered-with-mass-hold |
| three-pilot-kits | provisional-definitions-delivered |
| coupled-actuator-and-launch-search | model-pass-provisional |
| baghdad-pilot-delivery-and-finance | provisional-pilot-delivered |
| lifecycle-and-assurance | execution-hold |

## Inspect the connected evidence

- controlled-product: Counted equipment, allowance substitution, physical kit identities and bay access.
- coupled-actuators: Voltage/current/force/beam/sensing dynamics and adverse cases.
- launch-options: Independent energised sections, conductor and voltage lifetime-cost comparisons.
- pilot-corridor: Baghdad pilot access, provisional demand, full-line service and funding.
- candidate-finance: Exact counted actuator alternative, actual powered sections, full-line energy and funding sensitivities.
- resource-delivery: Resource-constrained1km erection, procurement and pooled factory accounting.
- native-lifecycle: Serial-specific simulated receiving, inspection, repair, replacement and handback.
- assurance-profiles: Separate executed engineering, delivery and formal control profiles.
- capability-migration: Actual v0.4 capability dispositions and three-product boundary.

Digital completion is recorded separately: 8 scoped digital subtasks completed; 576 broader qualification tasks tracked; 0 full qualification tasks completed and 0 physical acceptances. The structured study links every subtask to its qualification parents and exact packet receipt.

Actual module-force screen: passed; observed peak 14247.581406153944N against the declared 20000N study bound. A controller's requested-force clamp does not bound actual magnetic-force overshoot. This screen does not qualify the physical attachments.

Retained rejected experiment: 120-turn coupled run and its exact sources. Its matched failed-candidate financial sensitivity remains separately archived. A later candidate does not overwrite these failures.

The component-accounted product is separate from historical14.2/14.4t tare targets. Defined equipment replaces the same scope allowance; target overruns, rejected assembly paths and unselected redesigns remain visible. Passing file integrity cannot close a failed design screen.

The counted steel/higher-copper actuator alternative has its own financial sensitivity. Its exact winding, cooling, mass and powered-section configuration are priced together. The lower-cost reference case and independent launch search cannot supply an economic advantage for a different executed candidate.

The pilot demand calculation is a declared open-data/proxy experiment, not an observed patronage forecast. Fare revenue, DSCR and funding gaps are sensitivities, not a funding commitment. The comparison-only steel-wheel option has its own quantity bill and does not replace the primary maglev architecture.

The native business campaign records simulation assets only. Receiving, inspection and handback in that campaign establish software workflow behavior; no hardware is accepted or authorised.

Missing v0.4 functions remain named: . They are not silently retired or marked restored.

External closure inputs: qualified force/loss and material maps, measured mass/CG/inertia, observed OD and boarding counts, surveyed land/utility/ground conditions, supplier configurations and quotations, fire/accessibility/evacuation acceptance, lender and authority decisions, independent physical first-article tests.

Reproduce: `./osr design pilot-delivery`. Heavy evidence: `./osr assurance run engineering`, `./osr assurance run delivery-lifecycle`, `./osr assurance run control-formal`.

# Controlled provisional pilot product

docs/design-engine/ferrite-maglev/pilot-product/README.md

Configurations: 2; exact product definitions: 3.

The scope ledger replaces each covered allocation with its definition and retains only the unrepresented residual. Defined mass above its allocation increases the total; the growth reserve is retained. Cost rates remain unquoted allowances.

baghdad-30kwh-aw3: target tare 14,200 kg; reconciled tare 16,817.026 kg, AW3 28,817.026 kg. These overruns prevent adoption of the old mass targets. baghdad-50kwh-aw3: target tare 14,400 kg; reconciled tare 17,017.026 kg, AW3 29,017.026 kg. These overruns prevent adoption of the old mass targets.

The equipped-bay geometry includes retained cartridge/positive retention, three winding-pack candidates, paired sensing, disconnect interfaces, power/signal/earth route definitions and a removable cover. Straight supported extraction uses conservative swept AABBs against the assembled body/guideway; reported obstructions are holds, never waived collisions. Cable bends, qualified fittings, tool access, magnetic handling and installed qualification remain open.

The retained cartridge-only downward path has 48 conservative bound obstructions. The separately identified segmented-collector/detachable-elbow lateral candidate has 0; it adds four splice plates and16M8 bolts per bay,106.014kg/car if adopted. Those unqualified splices are not included in the base product mass or silently selected.

Dimensioned bay views · Calculated replacement animation · Exact translation events. These are code-native bounds, not native-detail connector or released manufacturing drawings.

No complete equipped mass, supplier quotation, magnetic map, manufacturing or physical release is claimed.

# Provisional electrical/magnetic/moving actuator composition

docs/design-engine/ferrite-maglev/pilot-actuators/README.md

The retained ideal-force benchmark remains separate. This new execution integrates actual public Rust current controllers with independently solved RL windings, gap-dependent support flux and forces, moving-body/guideway states and local powered-group phase references. Copper heat, DC energy, current/voltage/temperature/phase screens and rejected cases are retained.

Nominal executed 1415/2000 moving frames; numerical feasibility False; failure FAULT 36:Some(Current).

Counted repair steel-wall-004/equal-copper-80t10mm2: 29,064.079kg/car AW3; 2000/2000 frames; numerical pass False; failure None; current screen False. Counted repair steel-wall-004/equal-copper-100t8mm2: 29,064.079kg/car AW3; 2000/2000 frames; numerical pass False; failure None; current screen False. Counted repair steel-wall-004/higher-copper-140t20mm2: 30,382.351kg/car AW3; 140417/140417 frames; numerical pass True; failure None; current screen True.

Equal-copper rewinds preserve I²R and current density at equal ampere-turns; they cannot cure the thermal failure by relabelling a60A rating. The counted higher-copper candidate derives its700mm mean turn from300×200mm outer and120×80mm inner dimensions,140turns×20mm² in a75mm-deep pack; actual copper, resistance, pack fill and vehicle mass are recalculated. The detachable collector adds its separately derived106.01368kg/car of splice plates and bolts, with no cut-stock removal credit. Additional mount36kg/car and cooling120kg/car with an explicit3× continuous winding heat-rejection allocation are included. The3.75A/mm² baseline wire screen and separate60A drive screen remain unchanged. Coldplate flow/thermal resistance is an unqualified design requirement, and full-trip adiabatic temperature does not qualify continuous duty.

Counted higher-copper packaging · Exact candidate bay definition. Existing26.4t hardpoint/civil acceptance is not inherited: the new gross load requires re-evaluation.

Each group uses a flux-weighted virtual phase reference. The full3degree mechanical/index allowance remains ADDITIONAL to actual car-relative phase and position, timestamp, stator-registration and thermal uncertainties. Worst-case magnitudes are summed against the unchanged5degree local limit. Seventy independently controlled5m groups per direction count140converters at a two-direction station; the actual12m band/17.5m pitch leaves5.5m between bands. Whole-train5degree spread is a distinct historical observation. The adverse index error also enters the physical field and its unwanted normal force.

The192V/60A support power-stage alternative adds48kg/car of converter allocation plus6kg/car of observer electronics and explicit unquoted prices. An actual standalone Rust observer (bounded by-value ABI) uses the one1ms-frame delayed calibrated phase/current/gap model and computes normal force and pitch/roll wrench compensation; this changes current commands before the same actual Rust PI/RL plant, never applies an ideal mechanical force. Nominal and both0.9/1.1observer-scale full journeys must pass all unchanged screens, alongside nine exact-candidate faults. The1mm nominal coldplate/carrier stock clearance has no approved tolerance or physical fit acceptance.

Gap-field screens use an assumed25,000mm² pole area, while the winding inner aperture is9,600mm². No core neck/yoke area or validated flux split has been derived. A low gap B cannot establish core saturation acceptance; the magnetic geometry remains an explicit manufacturing/field-map hold. Updated body/beam gap poses are checked after every mechanical frame, including the final frame.

The nominal motion profile uses actual Baghdad line14 mainleg geometry, with runtime rounded upward to the1ms model clock; finance must use that exact profile hash. Rust current input power and heat are aggregate support/wayside values; they must not be added to a separate LSM loss budget as support-only energy. The RL energy identity covers frozen-geometry incremental current energy, not a complete measured PM/structure energy balance.

Assumed25,000mm² gap pole area versus9,600mm² winding aperture: pole shoe/core neck/yoke flux geometry and saturation remain unreconciled; gap-field pass does not close this hold Support/LSM maps, shared PM flux, saturation/core/eddy losses and temperature qualification Lateral/guidance dynamics and all omitted current circuits Complete converter/DC/brake/cooling/electrical fault energy and physical timing Whole equipped-body mass/CG/inertia and joined load paths remain unaccepted One-frame normal-wrench observer uses unqualified calibrated phase/current/gap maps; full0.9/1.0/1.1scale cases are required, not a physical robustness proof

No digital pass authorizes hardware, manufacturing or operation.

# Provisional finite-group launch optimisation

docs/design-engine/ferrite-maglev/launch-optimisation/README.md

108 equal-service winding/group/voltage candidates; 33 satisfy the implemented analytical current/voltage/heat screens.

Each result binds the exact component-accounted vehicle configuration and motion profile. Groups contain whole2.5m cassettes; the loss-minimizing finite-group current allocation is rejected when current or voltage limits cannot produce the requested trajectory. Booster stock and heat remain separately counted.

Installed lifetime cost includes finished copper, every converter and sensor/protection group, stator hardware, cooling, idle power, maintenance and converter renewal. Assumed train passages and regeneration reuse are explicit finance sensitivities, not route or grid acceptance.

No measured force/flux/inductance/saturation map or hardware-current qualification Optimal instantaneous sharing requires physical enable/ramp and protected handover execution Adiabatic winding temperature is not cooling/ambient duty acceptance Cost inputs are unquoted assumptions; converter/device/cabinet/current capability and switching loss curves unselected Grid absorption/recovery, protection/braking redundancy, service duty and financing remain separately qualified

No supplier quote, measured magnetic map, physical output, manufacturing release or default adoption.

# Resource-loaded Baghdad delivery study

docs/design-engine/ferrite-maglev/delivery-schedule/README.md

A complete finite 1 km task graph counts 80 directional beams, 41 support positions, finite mould/curing/QA/haul/crane/stator/test resources and procurement lead times. All inspections and strength certificates remain unexecuted.

Schedule and quantities · Scenario chart · Receipt

- nominal: 160.17 calendar days; 686 tasks.
- rework: 160.17 calendar days; 702 tasks.
- import-delay: 184.29 calendar days; 686 tasks.
- crane-outage: 160.17 calendar days; 687 tasks.
- slow-cure: 170.29 calendar days; 686 tasks.

Factory capital is a single100km programme pool with a route-km allocation, not a separate whole factory charged to every city. Imported finished equipment, imported inputs, local process expenditure and training are separated; local spend is not verified domestic value added. UK suppliers are research candidates; no ordered model, origin certificate or quote is claimed. See dated primary-source links in assumptions.

# Baghdad current-line pilot delivery study

docs/design-engine/ferrite-maglev/pilot-corridor/README.md

Proposed complete line14: four stops over6.042km, with both retained terminal loops. This is a source-bound planning package, not permission to build or run passengers.

Open-map access · Service diagram · Complete calculations/raw events · Source receipt

WorldPop2026 modelled20min walking catchment: 264,280 residents; the independent GHSL2025 model gives 95,066. This substantial source disagreement is not resolved by averaging. Residential pixels are counted once. Strict verified accessible coverage and employment/OD data remain unknown. Step-excluded OSM routes,150m unsurveyed connectors, wheeling speed and lift outages are explicit sensitivities. Dated WorldPop/Geofabrik/OSM primary data and attribution are retained.

Current240s minimum separation remains; finite four-hour service includes all12moving/terminal events per circuit. This does not demonstrate240s whole-line service: nominal maximum observed departure gap is 241.0s, including finite startup and receiving-berth constraints. Nominal fleet 10 including2spares. Failed receiving/charging sites, variable dwell, safe withdrawal and bounded recovery are separate complete replays. Python resource/energy replay is not the progressive Rust prototype or physical whole-line proof. Synthetic OD and segment capacity/unserved demand are exposed.

Unquoted pilot capital with reserve: $105,994,347; annual operating allowance $5,446,817. This reference budgets219m platforms. Unselected114.5m direction-aligned platforms produce a sensitivity of $103,006,190, conditional on a fitted504.5m power-zone and approaches; no compact saving is adopted. Fares, ramp-up, non-fare revenue, explicit subsidy, renewals, annual debt service/DSCR and funding gaps remain sensitivities, not financing commitments. One shared factory pool is allocated by programme output; no origin or local-value certificate is invented. Independent equal-service elevated steel-wheel deck/track/vehicle quantities are comparison-only. Shared rail foundation/column allowances are an unresolved scope hypothesis; heavier deck reactions require independent structural/ground design.

Aligned bodies shorten platform104.5m but opposite350m power zones require504.5m common straight; old400m fit cannot be reused. Same-level opposite-direction transfer has no lift; cross-line transfer needs separately fitted levels/core.

Remaining gates:

- actual passenger OD, mode share, jobs and affordability survey
- station entrances, kerbs, sidewalks, grades, crossings, lift evacuation/power and all site/land permissions
- full route and turnback curves, independent brakes, receiving/charging hardware and actual Rust multi-domain commissioning
- exact equipped product configuration/loss/price reconciliation, vendor quotes/origin, lenders and signed public support

# Separate counted actuator candidate: pilot cost and finance

docs/design-engine/ferrite-maglev/pilot-variants/README.md

Computed candidate and raw replays · Placed-part volume audit · Overview · Exact source receipt

Full coupled profile confirmed: True. Executed 140,417/140,417 moving steps; retained failure: None. All applicable full-profile estimator sensitivities passed: True. A completed planning DES or a nominal passing trajectory does not override a failed full-profile sensitivity.

Candidate unquoted capital $146,887,838; annual operating allowance $7,242,811. Exact component cost rows and procurement-linked capital cash, six service/fault replays and fresh debt/DSCR/funding gaps are retained.

The 192V support converter and observer equipment is counted separately: 54kg/car and $26,400/car, with 1kW/car of additional auxiliary allowance. Component rows reconcile to the exact tested configuration; the auxiliary allowance enters service energy once. Hardware, sensing, maps and timing remain unqualified.

Current candidate extraction video · Poster · Dimensioned stock bounds · Exact candidate definition · Translation events · Visual/source receipt. The larger140turn coil geometry comes from the exact canonical actuator candidate definition. This is a calculated conservative stock-bound straight-translation illustration, with the declared services/splices/retention released first; it is not measured motion or collision, lifting, magnetic handling or physical assembly acceptance. The source receipt binds retained hashes, candidate configuration, definition and events. Portable verification decodes all41frames and checks independently projected material footprints inside the viewport, with a bounded two-pixel edge allowance for lossy video. Font and encoder bytes are excluded from the oracle. The per-material check rejects a rehashed smaller-coil reference video; raster allowances are not engineering dimensional tolerances. Decoder metadata is checked first and output is capped at42fixed-size frames.

The original GFRP/50turn financial study is preserved. The candidate has independently counted steel tubes/spacers, additional winding copper, mounts and cooling. Costs carry no unidentified GFRP removal credit. Vehicle integration and design reserves apply to the component additions; no overall capital multiplier substitutes for a BOM.

Mainline winding/section/voltage choices are recomputed at the candidate mass and exact retained coupled trajectory. Full-duration actuator evidence must have the same configuration and profile; a short or different-profile pass cannot close that gate. Terminal-loop/guidance motion and physical hardware remain held.

The independent placed-box audit finds 0 inherited net-volume records above the actual placed gross dimensional bound. Segmented collector metadata requires reconciliation. Gross bounds are not net manufactured material; all net-solid/manufacturing quantities remain held. The conservative full-base stock plus added hardware mass/cost remains unchanged.

The opaque60kW/car hotel allowance remains. A separately calculated60A continuous winding $I^2R$ ceiling and counted cooling auxiliaries are added conservatively; overlap with that unitemised allowance receives no removal credit. Mixed support/wayside Rust heat is not added to the independently computed LSM DC import. This power ceiling is not measured duty or a cooling qualification.

All prices are planning inputs. Neither candidate nor base is adopted or released for manufacture, passengers, physical operation or financial close.

# Current maglev pilot lifecycle

docs/design-engine/ferrite-maglev/pilot-lifecycle/README.md

Current-maglev draft procurement, native business execution when separately retained, serial inspection/rework, installation, alarm/repair, replacement and independent software-business review. No commissioning or operating authority.

The three pilot positions use the exact controlled product/BOM revisions. Costs remain unquoted; stock transactions are explicit software fixtures. The journal rejects a late failed inspection/test, stale configuration, wrong role and self review. Replacement keeps the removed serial and original records. A proposed revision reopens business review and test holds.

Native stack evidence: executed-current. Use `tools/automation/pilot-lifecycle-native.py run` for an isolated disposable campaign and `verify` for retained evidence only. Native ERPNext owns purchasing, stock, quality, production and Asset Repair. FUXA observes simulated conditions only. The authenticated ERP Workbench exposes serial actions and business evidence. A business handback never changes physical/manufacturing release or movement authority. Independent physical acceptance remains open.

Physical gates: Measured magnetic/thermal maps and qualified cartridge process, Guidebeam materials, prestress, handling and site acceptance, Target controls/timing and integrated protection, Fire/evacuation and independent engineering/authority acceptance.

# Pilot development assurance profiles

docs/design-engine/ferrite-maglev/assurance-profiles/README.md

Exact development execution; engineering, delivery and formal profiles are separate. No SIL, hardware, site or operational acceptance.

| Profile | Digital execution | Commands |
|---|---|---|
| engineering | hold | 0 |
| delivery-lifecycle | hold | 0 |
| control-formal | executed-current | 5 |

Run an explicit profile with `./osr assurance run <profile>`. Opening/verifying this packet only checks retained evidence. Failed or stale runs cannot become a current pass. Build current design packets before an engineering profile; its pending hold is published before readiness tests. Run `./osr build` after successful profiles to refresh their completed presentation.

The project assurance graph traces current engineering, delivery and control claims to source. Source traceability, executed verification and independent acceptance are separate gates.

Physical gates: measured magnetic maps, target timing and EMI, site/ground and utilities, fire and evacuation, supplier qualifications, independent engineering and authority acceptance.

Compile the whole pilot evidence graph with `cargo run --locked -p osr-safety-case -- engineering/ferrite_maglev/data --evidence-root . --results docs/design-engine/ferrite-maglev/assurance-profiles/results.toml`. The physical-protocol branch remains unverified and every independent acceptance remains open.

Kani usage and harness selection · Loop bounds and unwinding assertions.

# v0.4 capability migration to station-powered maglev

docs/design-engine/ferrite-maglev/capability-migration/README.md

Migration of the actual v0.4 feature table. Tested scope is stated per row; missing features are not marked complete.

Baseline: v0.4.0 commit `205e8dc158c83fcdc2dca9ceb8bad0a7e8b3f61e`, README feature table. Historical product quantities and wheelset geometry are not current inputs.

| Capability | Disposition | Accountable function | Current evidence and limits |
|---|---|---|---|
| Deterministic city generation | retained-but-unconnected | City planning | Current266city generation retains primary geographical identities; a separate16layer stop/service/OD proposal workspace is now restored. cities.py |
| Generatable project digital twin | replaced | Delivery engineering | Generic any-city revision compiler now emits planned workfront/resource/procurement/monthlycash and retainedERPadapter inputs. Missing rates and actual native records remain explicit; no complete city delivery acceptance. delivery_platform.py; platform_publication.py |
| Integrated Workbench | replaced | Platform integration | Separate editable owner/builder GIS workspace, evidence viewer and authenticated native owner/builder/operator/people/observation navigation. CompleteOCC/HR federation remains open. maglev-workbench.py; planning.js; platform_publication.py |
| Business execution and embedded maintenance | retained-but-unconnected | Owner/operator platform | Test scope is serial cartridge and beam simulation in native ERPNext/FUXA; entire-city HR/accounts and evaluator bindings remain open. pilot_lifecycle.py |
| Interactive network and service planning | replaced | GIS and transport planning | 16named source-bound GIS layers; unavailable/sampled layers labelled. Stop/service/dwell/OD and straight-chord proposals, immutable revisions, stale/tampered refusal and local export/import are implemented. General alignment/OD calibration and engineering adoption remain open. delivery_platform.py; platform_publication.py; planning.js |
| Software in the loop | replaced | Control integration | ActualRust two-circuit support/guidance/current dynamics,realizablybounded preferred allocation,live numerical authority and postfault integration replace earlier detection-only coupons. Failed maps/capture/thermal/designcases remain failures; no physical acceptance. full_dynamics.py; candidate_authority.py |
| Independent operations cross-check | replaced | Operations modelling | Executed independentSUMO Railcustom timing/block/berth,dwell/receiverdelay and over-capacity controls use identical passenger stops and candidate mass. Kinematictraction is explicit; magnetic/Rust/HW and whole-fleet physical assurance are not inferred. operations_crosscheck.py |
| Civil BIM, GIS and field evidence | replaced | Civil information management | One magnetic module and25m concretebox in actualIFC4.3 with reopenedvolumes,executedIDS1staleidentityrefusal,BCF3schema-valid issues and typed unaccepted field readings. Wholecity IFC/physicalas-built federation remains open. bim_field.py |
| Station and civil product geometry | replaced | Station and civil engineering | Current guidebeam/hub definitions replace old platform and bogie-change products; all45former families are not reproduced. shared_hub_layout.py; pilot_product.py |
| Station engineering screens | replaced | Station engineering | Current structural/access screens replace old benchmarks; complete fire/drainage/thermal qualification remains open. station_native.py; hub_access.py |
| Buildable modular trainset | replaced | Mechanical product engineering | Current composite body and magnetic bays replace the historical wheel/traction product; stock geometry is not a manufactured train release. assembled_body.py; pilot_product.py |

| Capability | Disposition | Accountable function | Current evidence and limits |
|---|---|---|---|
| Automatic cost propagation | replaced | Project economics | Counted newcandidate quantities,cutyield/ prices,commonhotel assumptions and single calendarstandby reconciliation. No newpilotfinance adoption whilemandatorypropulsion/fullserviceduty fail; priorconfiguration finance remains separate. candidate_economics.py; candidate_manufacturing.py; launch_optimisation.py |
| Operations and assurance | retained-and-tested | Independent assurance | Executed authority and maglev refusal properties are source-bound; they do not provide physical or operational acceptance. pilot_assurance.py |
| Deterministic integrated testing | replaced | Platform verification | Current disposable native scenario exercises ERPNext/FUXA/Workbench with faults/recovery; broad former editor coverage is not restored. pilot-lifecycle-native.py |

# Three products

Owner/builder/operator delivery platform — Planning, procurement, simulated QA/serial execution, evidence and maintenance with business-owned permissions. Controlled pilot planning and training before rolling-stock certification; no movement authority.

Reference hardware and factories — Mechanical/civil candidates, methods, origin-accounted suppliers, QA and serial definitions. Digital pilot kits for review and supplier engagement; manufacture requires independent release.

Open GOA4 control platform — Rust authority/control kernels, supervision, offline plant and refusal tests. Replay and hardware-in-loop development; no live train or safety authorisation.

The first adoptable product is the simulated owner/builder/operator delivery workflow. It may be used for controlled planning and training before rolling-stock certification; legal entities, calendars, permissions and real equipment bindings remain deployment inputs. The read-only maglev evidence Workbench and the authenticated native lifecycle application have different responsibilities. Neither grants movement authority or accepts manufactured hardware.

# Detailed candidate and system simulation

docs/design-engine/ferrite-maglev/system-detailed/README.md

Completed bounded digital development and simulation publication, including rejected designs. A complete physically qualified train/civil/software system has NOT been reached.

Exact candidate `a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd`: 25.619t tare and 37.619t AW3 per car, six cars. Geometry-derived equipment is counted; the old94ad actuator pass is a separate configuration.

| Development chain | Result |
|---|---|
| magnetic-static-support | OPEN / FAILED: Actual thin return-rail flux cross-section and saturation; target AW3 support must be achievable within current/material limits |
| magnetic-propulsion | OPEN / FAILED: Independent applicable field/convergence/demagnetisation evidence, not the parent model |
| structural-and-interface | OPEN / FAILED: Actual body,attachment,guideway,handling and provisionalground screens |
| coupled-motion-and-faults | OPEN / FAILED: Actual currents,continued fault states,guidance and authority; no detection-only pass |
| native-candidate-delivery | Implemented screens passed: Exact quantitative products through source-bound native ERP/FUXA lifecycle |
| physical-and-manufacturing-acceptance | OPEN / FAILED: Measurements,approved drawings,materials,hardware timing and independent acceptance remain required |

## Engineering and delivery packets

- candidate-native — exact source receipt and retained results.
- candidate-magnetics — exact source receipt and retained results.
- candidate-structures — exact source receipt and retained results.
- full-dynamics — exact source receipt and retained results.
- candidate-manufacturing — exact source receipt and retained results.
- candidate-lifecycle — exact source receipt and retained results.
- bim-field — exact source receipt and retained results.
- operations-crosscheck — exact source receipt and retained results.
- planning-platform — exact source receipt and retained results.
- planning-ui — exact source receipt and retained results.

The candidate has a complete traced publication chain, not a completed engineering design. Failed field, structural or dynamic evidence blocks adoption even when BIM, purchasing or independent operations tests pass. No file or software-business result grants manufacturing or physical release.

The economics packet separates counted material/process subtotals from complete installed price. It compares identical passenger-services assumptions on a common kinematic trajectory. The new candidate has no adopted CAPEX/OPEX/debt forecast while mandatory propulsion and full-service duty evidence fail; the earlier configuration's finance is not relabelled.

The owner/builder planning workspace can be used for controlled proposals independently of train physical qualification. Its generic workfront schedule and native navigation do not imply whole-city BIM or full OCC/HR deployment.

Remaining engineering work:

- Static lift: actual25mm return-rail flux area saturates before required AW3 support.
- Propulsion topology and refined demagnetisation/convergence.
- Joined carbody/splices,braking/capture interfaces and fatigue/crash.
- Rigid magnetic array versus terminal curvature and route recovery.
- Full four-stop motion/dwell/charging and fleet dispatch with qualified useful-field maps.
- Detailed electrical protection/PCB/insulation/thermal/hardware timing.
- Geotechnical foundation,station/depot/fire/evacuation and physical construction stages.
- Complete installed cost,local origin,verified demand and financing commitments.

Reproduce: `./osr design detailed-system` after the component packets. `./osr build` regenerates stale implemented studies; it does not autonomously produce physical qualification.

# Dimensioned magnetic circuits and passive propulsion alternatives

docs/design-engine/ferrite-maglev/candidate-magnetics/README.md

Provisional common candidate a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd. Parent94ad remains historical evidence.

Counted tare 25618.592 kg/car; AW3 37618.592 kg/car, six cars. Only exact old cartridge/winding scopes are removed. New cores, ferrite, windings, cradle, protection and propulsion stock are counted.

Lift/guidance: four actual air gaps, shared nonlinear steel stem, two PM branches and three physical coils/two electrical circuits. Typical manufacturer material curves, PM recoil/operating points, differential mutual inductance, force/coenergy reciprocity and conditional repeated cooling are implemented. Independent complete3D lift/guidance maps remain open.

Propulsion: executed independent2D magnetostatic coupons retain Halbach, rectangular-shoe and round-PM alternatives. The corrected25mmreturn-rail cross-section prevents the assumed AW3 support; no propulsion alternative passes all applicable field/demagnetization/convergence screens. The group map returns conditional undamaged-field diagnostics and refuses adoption. New slot/turn/current/copper/thermal quantities are counted; phase end routing and impedance remain unqualified.

Common geometry/mass/CG/inertia · Executed studies and failures · Module · Propulsion band · Assembly illustration · Cut/material/process BOM · Receipt

Core curve: SIJ. Ferrite: TDK FB9B/recoil guide and FB12H. UK cut/stack and winding candidates: H V Wooding, Brandauer, Trans-Tronic. UK manufacture of those processes does not establish British origin of steel/ferrite/conductor or a supplier quotation.

No manufacturing, engineering, physical, site or operating release is claimed.

# Exact candidate native module

docs/design-engine/ferrite-maglev/candidate-native/README.md

FreeCAD and STEP occupied volumes/centroids recover the same module mass from declared densities. Existing exact same-definition CAD/STEP and their reopen receipts are retained unchanged when their exporter/source hashes agree.

Retained native triangles are independently checked for finite nonempty closed oriented topology, exact source/void surface membership, bounds, signed volume and centroid. Tessellation volume tolerance is1%; BRep remains1e-7. The deterministic orthographic image is checked pixel-for-pixel in its geometry viewport. All96 decoded frames of the four-second/four-stage video are checked against native geometry masks/material colours; fonts and encoder bytes are excluded from semantic matching.

The clip is final-position part visibility, not motion, collision, lifting or assembly-procedure simulation. Complete wire routing, cooling, connectors, fit and physical qualification remain open. No manufacturing or physical release is claimed.

Image · Assembly visibility video · FreeCAD · STEP · Geometry audit

# Candidate-specific structural design study

docs/design-engine/ferrite-maglev/candidate-structures/README.md

Exact candidate `a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd`; 37618.592 kg AW3 per car. All physical, engineering and manufacturing release flags remain false.

This package recalculates full force/moment equilibrium from the counted installed-mass placements. It does not scale the earlier26.4t interface result or adopt historic94ad dynamics.

Implemented results: hardpoint utilisation 0.568; retained M8 segment-splice utilisation 14.251; two-spine-only dynamic bending stress 589.5 MPa. Failed controls remain recorded; deeper collector/spine alternatives require a new shared candidate identity and repeated mass/fit/dynamic checks.

Reference25m span: 18.056 m³ concrete, 14946.1 kg new L-return/support steel, 1202.7 kg reinforcement, 604.5 kg strand, 61893.3 kg gross. This includes counted actual reaction rails, not the old7.51t allowance.

The original hypothetical60t@25m crane remains rejected. A separately identifiedLTM1750 manufacturer chart is evaluated at its26m row,32.7m boom and204t ballast;2.6t hook and3t slings/spreader are counted. Actual crane configuration, wind, access, ground reactions/mats and lift plan remain unapproved.

Material interference witnesses: 0. Zero witnesses alone does not qualify swept clearance or tolerances. New collector drilling is explicit machining with zero mass-removal credit. Concrete around fixed support embeds/blockouts is gross conservative stock, not completed net manufactured solids.

Soil cases50/100/150kPa and pile shaft/toe values are analyst sensitivities. No bearing resistance, groundwater, settlement or seismic acceptance is inferred from SoilGrids/open maps. Foundations and pier alternatives are separate, not additive.

Section drawing · Geometry image · Candidate load plan · BOM · Full study · Applied-load export · Counted geometry

Publication verification recomputes the structural study and independently checks retained payloads, exact source hashes, geometry-driven SVG and decoded section-image pixels. Image container bytes/fonts are not an engineering oracle.

QA and sequence records are unexecuted plans. No manufacturer quotation, concrete design code compliance, physical trial or site-specific design acceptance is claimed.

Primary references:

- JRC68415 — European Commission2012 bridge worked examples: separate permanent/live/wind/fatigue/construction actions, concrete, foundations and seismic. Road-bridge examples; not a maglev or Iraq compliance route.
- FHWA-PSC — Primary worked-example structure: transfer/service stresses, losses, flexure/shear, bearings, pier and footing. Our SI material limits/factors are explicitly unqualified inputs, not imported AASHTO acceptance.
- WSDOT-handling-2017 — Handling/shipping stages, transverse sweep/tilt, transport support parameters and strength at release must be stated; our simple roll/bending screens do not implement full PCI lateral-stability design.
- BCSA51-10 — UK steelwork fabrication, bolting/tool access, preloaded/slip-resistant bridge joints, weld procedures and fatigue. Historical2010 guide, no current code compliance claim.
- RSSB-GMRT2100 — Primary/secondary rail vehicle structure and passive-safety scope. No GB category or EN12663 load case is claimed; full standard and Iraqi applicability need independent review.
- ISRIC-SoilGrids — Global pedological prediction maps with uncertainty; no bearing, pile resistance, groundwater, liquefaction or settlement values inferred. All50/100/150kPa scenarios are analyst hypotheses.
- Liebherr-crane-scope — Manufacturer300t maximum is at3m radius and configurations vary. This package uses explicitly hypothetical radius-specific capacities and retains signed lift-chart/ground approval as a hold.
- LTM1750-9.1 manufacturer technical data, printed pages8/26, accessed2026-10-23.

# Continued six-car fault dynamics

docs/design-engine/ferrite-maglev/full-dynamics/README.md

Source-bound continued-fault digital study. Every actual terminal outcome, including failed capture, timeout and geometric/map hold, is retained. No output disappearance is classified as a safe stop.

Exact common candidate: `a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd`. AW3 mass per car: 37,618.592 kg. Equipment-derived CG and full inertia tensor include named reserve/capture/brake allocations once.

| Case | Physical-model terminal state | Continued post-fault frames | Implemented screens |
|---|---|---|---|
| straight-supported-stop | hold-initial-support-current-or-field-capacity | 0 | False |
| peak-current-main-power-loss | hold-initial-support-current-or-field-capacity | 0 | False |
| midspan-main-power-loss | hold-initial-support-current-or-field-capacity | 0 | False |
| handover-inverter-loss | hold-initial-support-current-or-field-capacity | 0 | False |
| receiver-braking-loss | hold-initial-support-current-or-field-capacity | 0 | False |
| sensing-disagreement | hold-initial-support-current-or-field-capacity | 0 | False |
| sensing-dropout-independent-reserve | hold-initial-support-current-or-field-capacity | 0 | False |
| watchdog-stale-map | hold-initial-support-current-or-field-capacity | 0 | False |
| nonuniform-map-errors | hold-initial-support-current-or-field-capacity | 0 | False |
| reserve-depletion-capture | hold-initial-support-current-or-field-capacity | 0 | False |
| no-capture-negative-control | hold-initial-support-current-or-field-capacity | 0 | False |
| asymmetric-load | hold-initial-support-current-or-field-capacity | 0 | False |
| urban-curve | hold-initial-curve-or-gap-geometry | 0 | False |
| terminal-loop-low-speed | hold-initial-curve-or-gap-geometry | 0 | False |

Every coil is integrated after a controller refusal: current decays in the explicitly declared freewheel path; permanent-magnet and mutual-circuit forces remain in the plant. A separate reserve controller and power/energy state are modelled. Counted unilateral shoes and friction clamps have declared, unqualified capacities, heat and travel limits. Detection alone never passes a stopping test.

The actual Rust allocation/current kernels provide bounded current demands and bridge voltages. Python integrates the nonlinear shared magnetic network and rigid-body/guideway states; there is no ideal-force substitution. The stationary1kHz/20kHz clocks and host execution observations do not establish target timing.

The six-body small-angle modes include vertical/lateral translation and roll/pitch/yaw, equipment-derived full inertia, gyroscopic terms, measured-state longitudinal spring/damper coupling, curvature chord errors, per-module asymmetry and per-span vertical/lateral/torsional modes. A rigid band does not automatically follow a tight curve. Guideway deadweight reactions include all counted stock; live modes reference dead-load-cambered geometry. Nonlinear magnetic energy, RL work, modal work, friction and reserve-energy balances are retained separately. Material hysteresis, nonlinear soil/joints and omitted shell modes remain open.

The actual Rust research allocator starts from source-bound minimax body-load preferences and enforces current-map residual-lift minimum/maximum forces, wrench and slew constraints. Static LP bounds iterate against actual six-car/guideway pole gaps. The previous zero-lift preference is not realizable merely because its force balance passes: permanent magnets still lift and cancellation can violate current/demagnetization limits. Full current inversion and actual bridge/RL dynamics execute every mechanical frame.

The receiver dq coupon is explicitly hypothetical. Its single .12Wb/4kA equivalent circuit cannot supply20kN: the actual Rust current guard refuses that demand. It must not be represented as a validated70group receiver or as the new magnetic geometry's useful external field. Shared-core leakage/no-map and dedicated passive propulsion-band alternatives require their own geometry-derived, numerically controlled maps.

A real source-bound Rust authority producer prepares the complete four-stop fixed line and samples actual front/tail poses. Its decision is composed with the actual140group stationary-alpha-beta current kernel. The refined magnetic-material failure blocks launch: full_trip_executed is false. This is an executed refusal attempt, not completed travel, dwell, charging, terminal turnaround or240s whole-line service. Portable raw protocol records are in authority/receipt.json. A controller/state-machine prototype for actual-pose dwell/energy/charge holds is present; it has not completed the full-line electromechanical trip.

Protection contact revision2 counts the named capture pads, steel hanger kits, brake hangers and paired transverse ties in the same vehicle mass/CG/inertia. Two12×200mm wear strips per track add37.68kg/m to the guideway. Contact-plane, magnetic influence, weld/bolt/rail-anchor fit, actuator travel and physical qualification remain holds. In particular,8×50kN pads cannot absorb a whole-car5mm drop within8mm at the present AW3 without sufficient residual/active magnetic support; a failed-capture outcome is preserved. Conditional numerical stopping results never remove these holds.

Selected development hardware: Raspberry Pi CM5 supervisory SBC and STM32H753XI local MCU. Dual sensing, sequence/configuration/time identity, independent reserve path, watchdog and isolated gate/protection interfaces are specified in interface-spec.json; physical adapters and target deadline measurements remain unexecuted.

Open gates: nonlinear map physical correlation, power stages and shared reserve independence, receiver field/phase and distributed-group compatibility, complete equipment fit and current-density/cooling duty, nonlinear joints/bearings/soil and flexible body modes, full-line protocol/berth/charge composition, target MCU/SBC timing/EMI and protection hardware, independent engineering and physical acceptance.

- stm32h753: Selected candidate MCU: Cortex-M7, ADC/PWM/watchdog resources; no measured railway execution claim.
- stm32-independent-watchdog: Independent watchdog uses separate low-speed clock; actual timeout tolerance and reset paths need target tests.
- cm5: Selected supervisory SBC: CM5, thermal throttling and published operating range; no fast-loop guarantee.

Reproduce explicit study: `tools/automation/osr-python -m engineering.ferrite_maglev.full_dynamics study`. Opening a retained packet never starts a simulation or hardware command.

# Quantified candidate manufacturing

docs/design-engine/ferrite-maglev/candidate-manufacturing/README.md

Exact candidate `a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd`. Three bounded products: magnetic module,25m guidebeam and5m wayside winding group.

Net material, purchase quantities, cut yield, waste, units, unquoted rates, valuation, lead-time hypotheses, source part IDs and process/inspection holds are preserved. Native BOM grouping sums constituent quantities; explicit native stock/currency quantisation preserves the exact engineering schedule separately; it does not substitute one USD1 kit for each part.

The launch coil is a dimensioned hypothesis. Failed or unresolved propulsion maps do not become qualified by purchasing it. Connector/cooling envelopes, fixture/tooling details, qualified tolerances, full factory labour and physical tests remain open. These are draft software-business inputs, not purchase instructions.

# Quantified candidate native lifecycle

docs/design-engine/ferrite-maglev/candidate-lifecycle/README.md

Candidate `a79a21a0767355fcb6518471769e532a04e2e23f4bd33db90968ef4a385df5bd`; native evidence executed-current.

The explicit runner uses the retained authenticated ERPNext/FUXA lifecycle for the three new candidate products. Native BOM/purchase/receipt quantities, stock units, unquoted rates and simulated asset valuation must match the engineering cut-yield schedule. Original/replacement serials, failed QA/rework, independent business review, process-loss recovery and scoped permissions remain required.

Run `tools/automation/osr-python tools/automation/pilot-lifecycle-native.py run --candidate`. Verification does not contact services. The earlier base-product campaign remains separate. Neither simulated dimensional measurements nor submitted business records approve the engineering design or physical railway.

# Candidate BIM and field exchange

docs/design-engine/ferrite-maglev/bim-field/README.md

IFC4.3 ADD2 actual component geometry, materials, configuration/part/serial identities and reopened boolean volumes. The executed IDS1 validator requires these identities; a deliberately stale component fails. BCF3 XML members validate against retained buildingSMART schemas and refer to actual IFC GUIDs.

The typed field adapter executes synthetic readings and rejects wrong units, out-of-range values, stale revisions and automatic acceptance. Every result awaits independent inspection. The7.5–8.5mm gap band is a provisional exercise limit, not an approved manufacturing tolerance.

Scope: one magnetic module and selected 25m narrow-deep-reference beam with counted concrete endinfills. Return rails and their support stock remain in the structural packet. Reinforcement/strand native detailing, station/depot, whole-city federation and physical as-built validation remain further work.

IfcTester, BCF3 specification.

# Independent operations cross-check

docs/design-engine/ferrite-maglev/operations-crosscheck/README.md

Actual Eclipse SUMO executes the same four passenger stops, six-car length and current candidate mass in both directions. Nominal, delayed dwell, receiver-unavailable berth delay and deliberately overbooked headway cases retain raw network, routes, stops, logs and output.

All declared kinematic/block screens passed: True. The nominal comparison uses a separate closed-form acceleration/cruise/braking timing oracle with a fixed5s comparison tolerance. Differences are reported, never calibrated away.

SUMO uses a declared constant-acceleration traction hypothesis for this independent scheduling comparison. It does not evaluate fixed-zone LSM thrust, magnetic suspension, independent capture, hardware braking, charging or actual Rust authority. Those are separate coupled system evaluators. Receiver unavailability here is a berth delay, not a magnetic stopping test.

Eclipse SUMO railway models and custom dynamics Eclipse SUMO installation and Python packages

No physical or operational acceptance. Reproduce with `./osr design operations-crosscheck`.

# Owner/builder planning and delivery platform

docs/design-engine/ferrite-maglev/planning-platform/README.md

266 current city revisions,16 named GIS layers, alignment/station/service/OD edit proposals, immutable local revisions and a generic resource/procurement/monthly-cash adapter.

Missing source layers are disabled. Streets and soil display samples are labelled;1km population aggregation preserves source population totals. Radial circles are display aids, not a walking catchment calculation. Editing a station invalidates access, demand, engineering and delivery assumptions until those are rerun.

The generic compiler executes the retained ERP planning adapter. It generates draft quantities and constrained workfront tasks for any retained city. Registered missing rates stay unknown; neither a priced subtotal nor a schedule is a complete project or released design. Native records and independent physical field acceptance remain separate.

Open the planning workspace. Export edits, then run `./osr plan compile edits.json --output build/my-plan`. A stale base revision or changed station order is refused. Local revision storage is append-only and recoverable.

Native owner, builder, operator, workforce and observation links use the isolated loopback stack and its existing authentication. The people link requires the relevant ERP module. This does not claim full OCC/HR federation or deployment permissions.

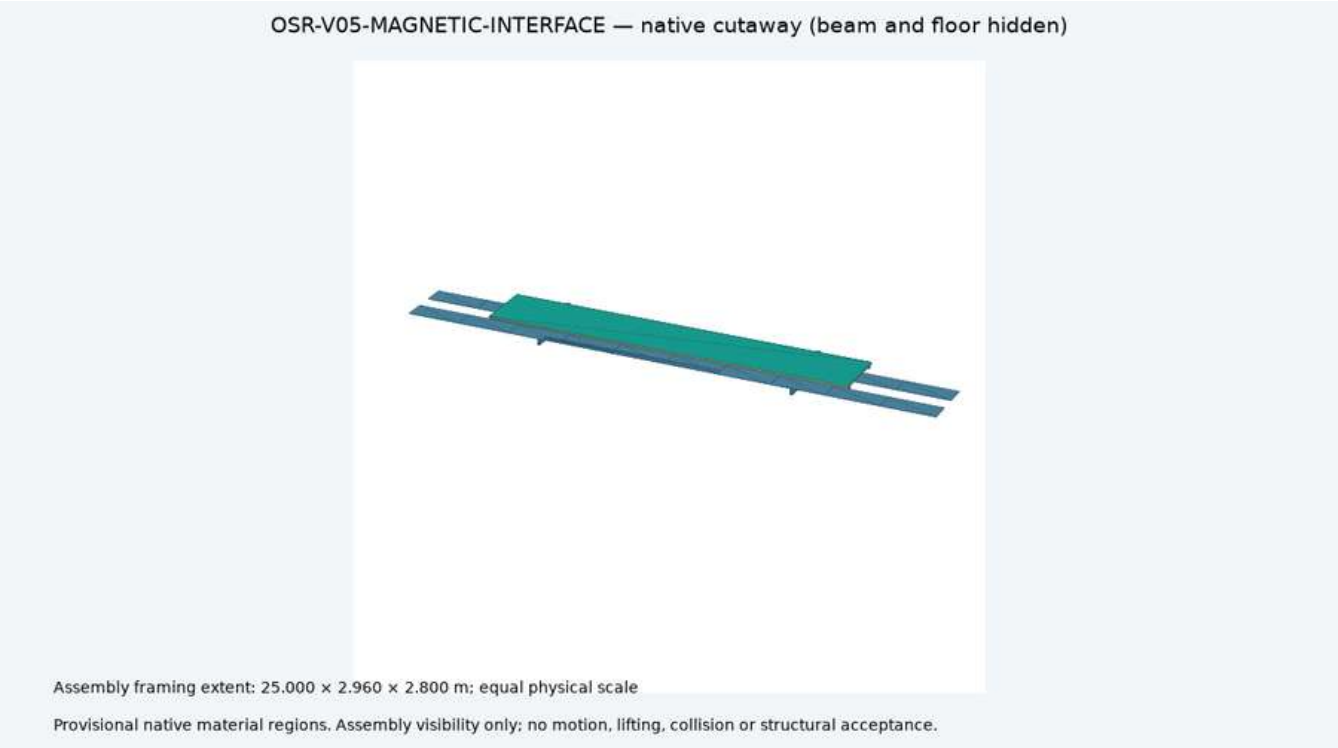
Primary geography and population sources remain in each city input receipt. All design, physical and procurement release flags remain false.

# Native geometry review




Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.

Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.



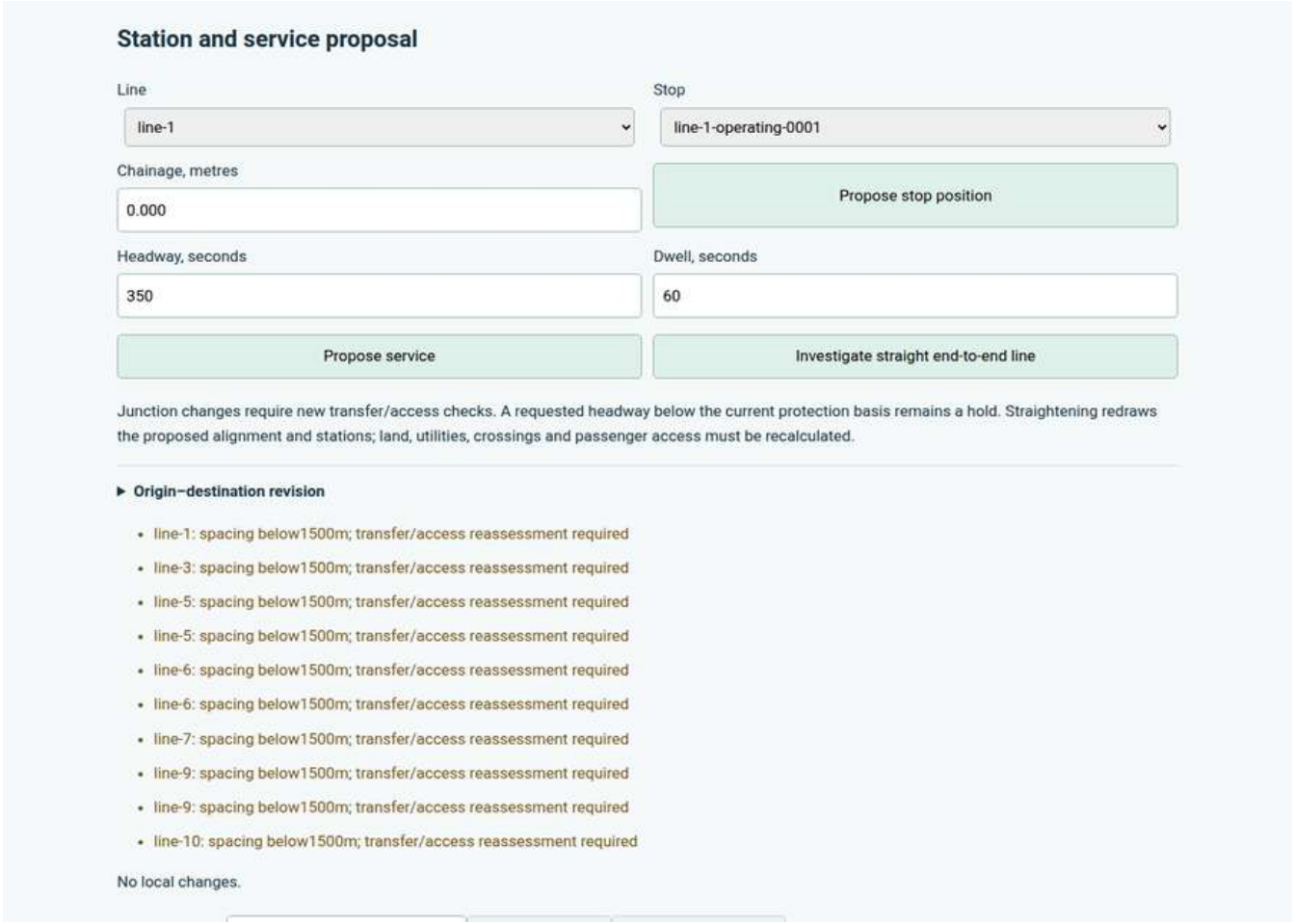

Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



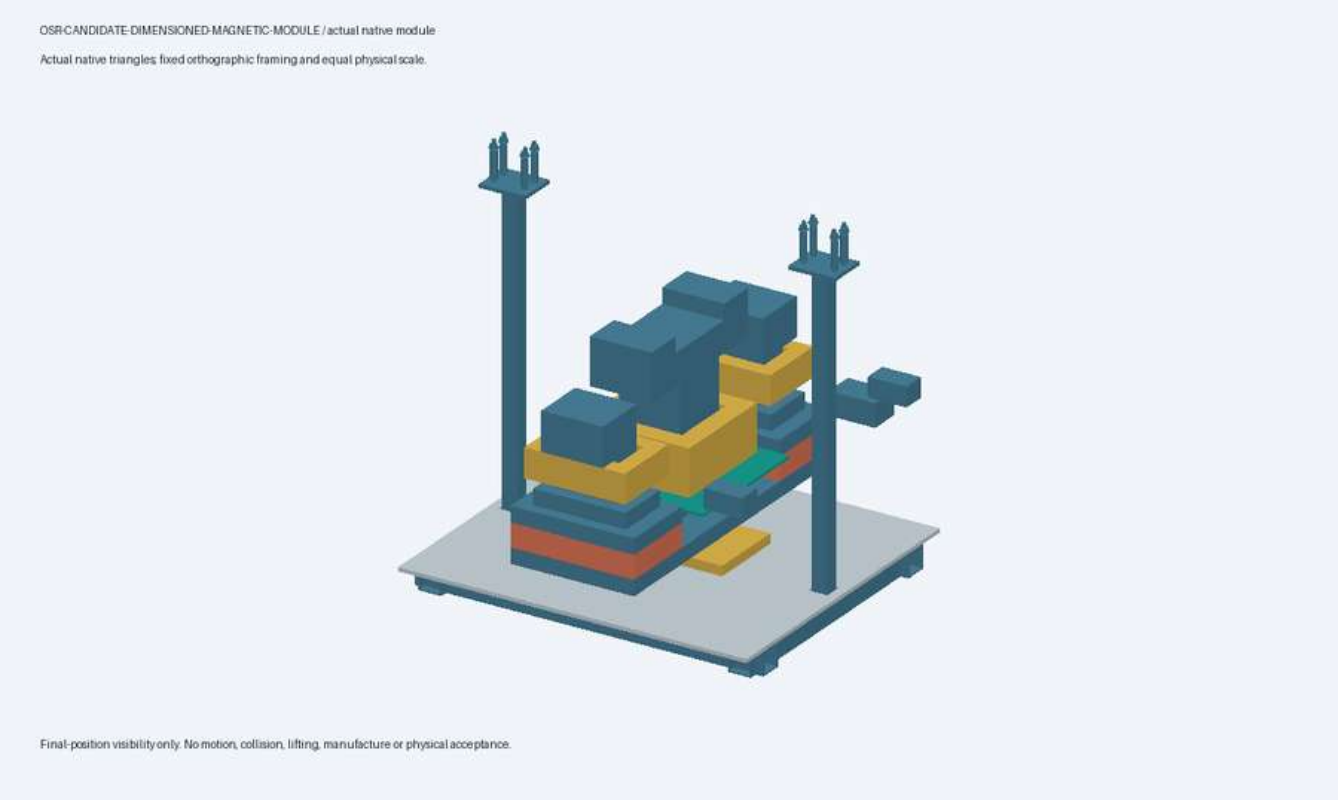

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.



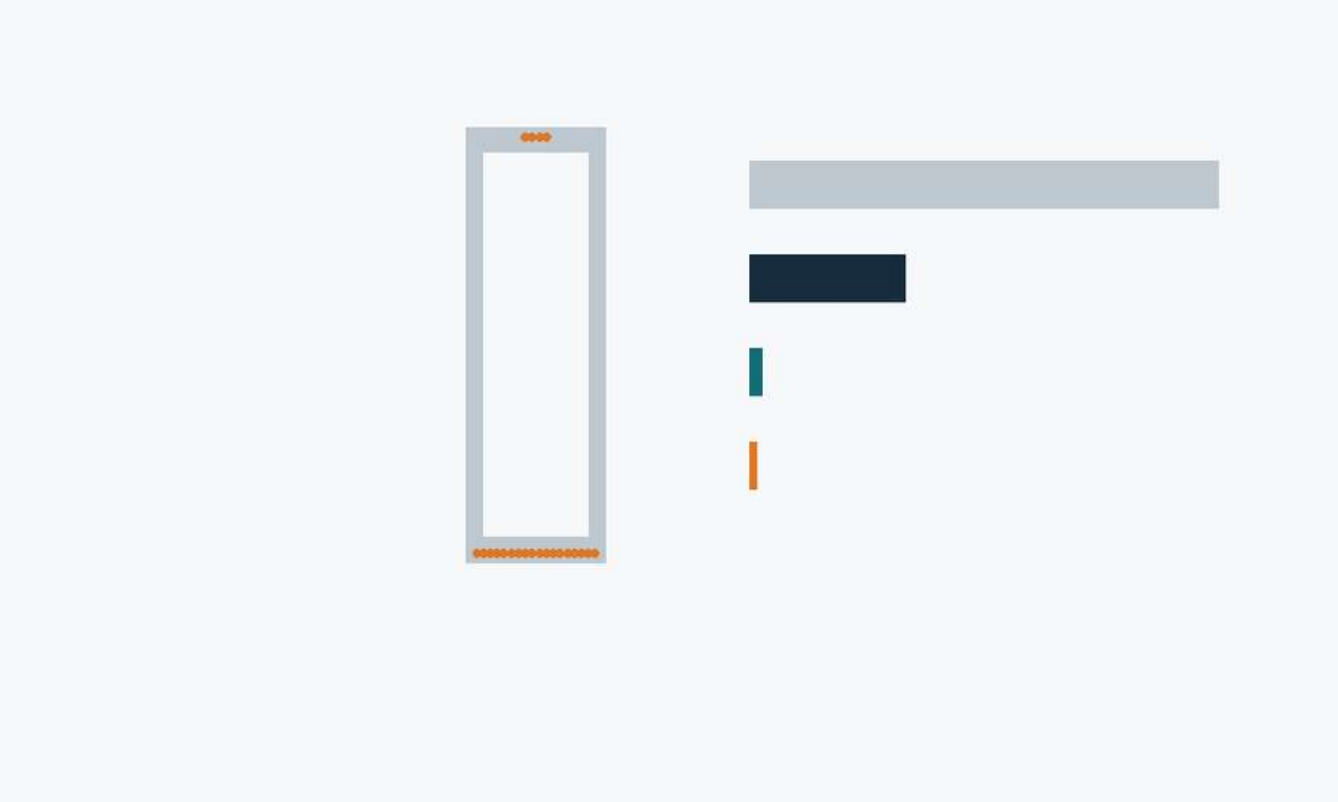

Actual editable owner/builder planning workspace; local proposals are not accepted routes.

Actual editable owner/builder planning workspace; local proposals are not accepted routes.




Exact dimensioned magnetic-module native solids; effective winding envelopes and physical qualification remain open.

Exact dimensioned magnetic-module native solids; effective winding envelopes and physical qualification remain open.



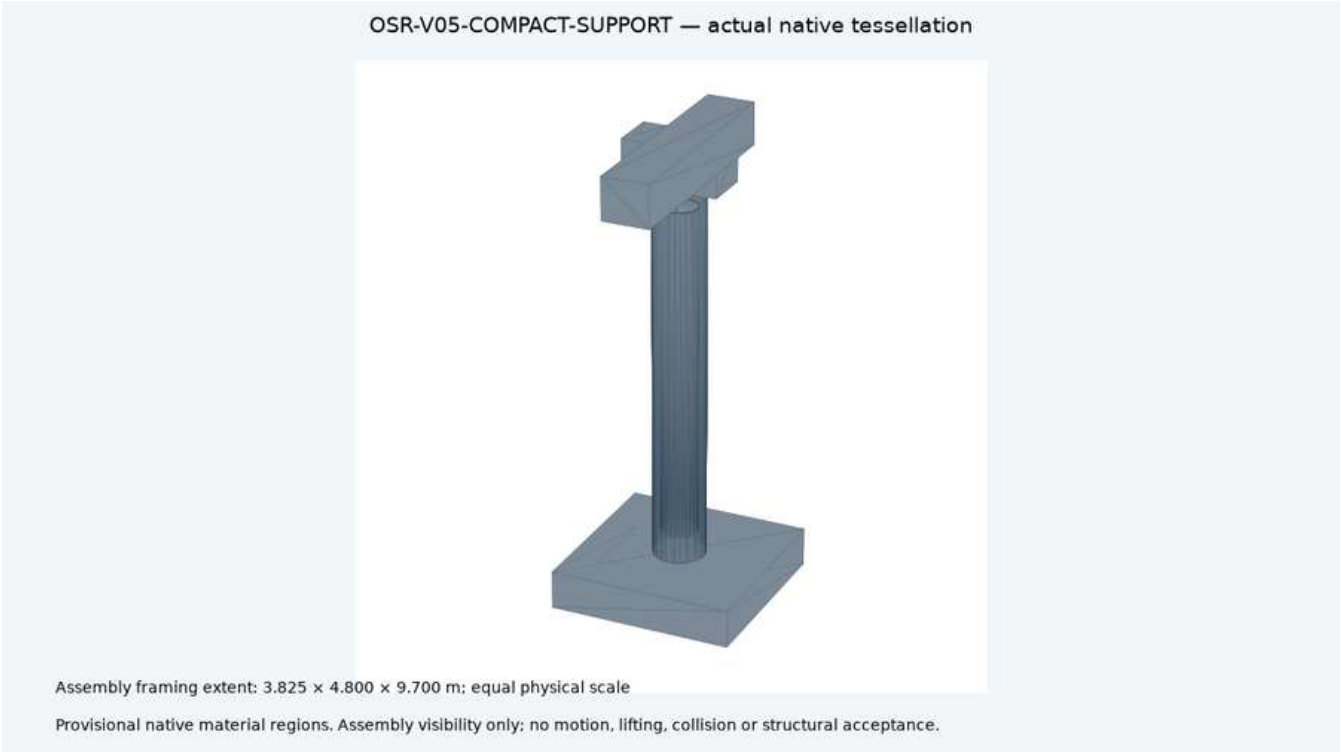
Candidate counted guidebeam section and prestress coordinates; no structural or site acceptance.

Candidate counted guidebeam section and prestress coordinates; no structural or site acceptance.




Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.




Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

# Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Baghdad | 451 | 2706 | 12 |
| Benguela | 87 | 261 | 12 |
| Huambo | 79 | 237 | 12 |
| Lobito | 57 | 171 | 12 |
| Luanda | 436 | 2616 | 12 |
| Lubango | 75 | 225 | 12 |
| Malanje | 24 | 72 | 12 |
| Namibe | 66 | 132 | 12 |
| Soyo | 51 | 102 | 12 |
| Uige | 16 | 48 | 12 |
| Bukavu | 89 | 267 | 12 |
| Goma | 72 | 216 | 12 |
| Kananga | 45 | 180 | 12 |
| Kinshasa | 600 | 3600 | 12 |
| Kisangani | 47 | 188 | 12 |
| Lubumbashi | 168 | 672 | 12 |
| Mbuji Mayi | 124 | 496 | 12 |
| Eldoret | 94 | 282 | 12 |
| Garissa | 49 | 98 | 12 |
| Kakamega | 61 | 122 | 12 |
| Kisii | 45 | 90 | 12 |
| Kisumu | 74 | 222 | 12 |
| Kitale | 63 | 126 | 12 |
| Machakos | 49 | 98 | 12 |
| Malindi | 49 | 98 | 12 |
| Meru Ke | 53 | 106 | 12 |
| Mombasa | 254 | 1016 | 12 |
| Nairobi | 540 | 3240 | 12 |
| Naivasha | 60 | 120 | 12 |
| Nakuru | 81 | 243 | 12 |
| Nyeri | 55 | 110 | 12 |
| Thika | 95 | 285 | 12 |
| Antananarivo | 347 | 2082 | 12 |
| Beira | 66 | 198 | 12 |
| Chimoio | 50 | 150 | 12 |
| Lichinga | 26 | 52 | 12 |
| Maputo | 204 | 816 | 12 |
| Nacala | 68 | 136 | 12 |
| Nampula | 75 | 225 | 12 |
| Pemba Mz | 54 | 108 | 12 |
| Quelimane | 15 | 45 | 12 |
| Tete | 69 | 207 | 12 |
| Xai Xai | 41 | 82 | 12 |
| Huye | 67 | 134 | 12 |
| Kigali | 186 | 744 | 12 |
| Rubavu | 62 | 124 | 12 |
| Mogadishu | 143 | 572 | 12 |
| Waw | 35 | 70 | 12 |
| Arusha | 80 | 240 | 12 |
| Dar Es Salaam | 523 | 3138 | 12 |
| Dodoma | 75 | 225 | 12 |
| Iringa | 51 | 102 | 12 |
| Kigoma | 49 | 98 | 12 |
| Mbeya | 69 | 207 | 12 |
| Morogoro | 68 | 204 | 12 |
| Moshi | 53 | 106 | 12 |
| Mwanza | 209 | 836 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Shinyanga | 55 | 110 | 12 |
| Songea | 21 | 42 | 12 |
| Sumbawanga | 30 | 60 | 12 |
| Tabora | 45 | 90 | 12 |
| Tanga | 70 | 210 | 12 |
| Zanzibar City | 81 | 243 | 12 |
| Arua | 56 | 112 | 12 |
| Entebbe | 65 | 130 | 12 |
| Fort Portal | 57 | 114 | 12 |
| Gulu | 96 | 288 | 12 |
| Hoima | 51 | 102 | 12 |
| Jinja | 76 | 152 | 12 |
| Kampala | 242 | 968 | 12 |
| Lira | 66 | 132 | 12 |
| Masaka | 52 | 104 | 12 |
| Mbale | 66 | 132 | 12 |
| Mbarara | 76 | 228 | 12 |
| Soroti | 22 | 44 | 12 |
| Lusaka | 284 | 1704 | 12 |
| Lyon | 279 | 1116 | 12 |
| La Paz | 232 | 928 | 12 |
| Cuenca | 93 | 279 | 12 |
| San Salvador | 266 | 1064 | 12 |
| Agadir | 112 | 336 | 12 |
| Beni Mellal | 44 | 88 | 12 |
| Fez | 137 | 548 | 12 |
| Kenitra | 81 | 243 | 12 |
| Khouribga | 29 | 58 | 12 |
| Marrakech | 221 | 884 | 12 |
| Meknes | 65 | 195 | 12 |
| Nador | 57 | 114 | 12 |
| Oujda | 50 | 150 | 12 |
| Safi | 59 | 177 | 12 |
| Tangier | 217 | 868 | 12 |
| Tetouan | 79 | 237 | 12 |
| El Obeid | 76 | 228 | 12 |
| Kassala | 47 | 141 | 12 |
| Khartoum | 435 | 2610 | 12 |
| Nyala | 67 | 201 | 12 |
| Omdurman | 290 | 1160 | 12 |
| Port Sudan | 54 | 162 | 12 |
| Tunis | 247 | 988 | 12 |
| Bloemfontein | 111 | 333 | 12 |
| Durban | 417 | 2502 | 12 |
| East London Za | 85 | 255 | 12 |
| Nelspruit | 54 | 108 | 12 |
| Polokwane | 78 | 234 | 12 |
| Herat | 59 | 177 | 12 |
| Jalalabad Af | 67 | 201 | 12 |
| Kabul | 254 | 1524 | 12 |
| Kandahar | 76 | 228 | 12 |
| Mazar E Sharif | 90 | 270 | 12 |
| Barisal | 95 | 285 | 12 |
| Chittagong | 354 | 2124 | 12 |
| Comilla | 95 | 285 | 12 |
| Gazipur | 269 | 1076 | 12 |
| Khulna | 226 | 904 | 12 |
| Mymensingh | 72 | 216 | 12 |
| Narayanganj | 102 | 306 | 12 |
| Rajshahi | 62 | 186 | 12 |
| Rangpur | 73 | 219 | 12 |
| Sylhet | 76 | 228 | 12 |
| Agra | 144 | 576 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Bhopal | 180 | 720 | 12 |
| Coimbatore | 314 | 1884 | 12 |
| Indore | 310 | 1860 | 12 |
| Jodhpur | 142 | 568 | 12 |
| Kanpur | 304 | 1824 | 12 |
| Lucknow | 309 | 1854 | 12 |
| Madurai | 205 | 820 | 12 |
| Meerut | 126 | 504 | 12 |
| Patna | 250 | 1000 | 12 |
| Raipur | 136 | 544 | 12 |
| Rajkot | 114 | 456 | 12 |
| Ranchi | 210 | 840 | 12 |
| Vadodara | 149 | 596 | 12 |
| Varanasi | 196 | 784 | 12 |
| Vijayawada | 184 | 736 | 12 |
| Visakhapatnam | 224 | 896 | 12 |
| Biratnagar | 54 | 108 | 12 |
| Kathmandu | 235 | 940 | 12 |
| Pokhara | 109 | 327 | 12 |
| Bahawalpur | 69 | 207 | 12 |
| Faisalabad | 181 | 1086 | 12 |
| Gujranwala | 202 | 808 | 12 |
| Hyderabad Pk | 216 | 864 | 12 |
| Karachi | 508 | 3048 | 12 |
| Larkana | 48 | 144 | 12 |
| Multan | 142 | 568 | 12 |
| Peshawar | 193 | 772 | 12 |
| Quetta | 124 | 496 | 12 |
| Rahim Yar Khan | 84 | 252 | 12 |
| Sheikhupura | 23 | 69 | 12 |
| Sialkot | 77 | 231 | 12 |
| Sukkur | 65 | 195 | 12 |
| Colombo | 345 | 2070 | 12 |
| Galle | 94 | 282 | 12 |
| Jaffna | 86 | 258 | 12 |
| Kandy | 97 | 291 | 12 |
| Phnom Penh | 280 | 1120 | 12 |
| Bandung | 243 | 972 | 12 |
| Surabaya | 284 | 1704 | 12 |
| Vientiane | 100 | 300 | 12 |
| Mandalay | 237 | 948 | 12 |
| Yangon | 474 | 2844 | 12 |
| Davao | 268 | 1072 | 12 |
| Ouagadougou | 227 | 908 | 12 |
| Bafoussam | 89 | 267 | 12 |
| Bamenda | 68 | 204 | 12 |
| Bertoua | 32 | 96 | 12 |
| Douala | 202 | 1212 | 12 |
| Edea | 12 | 24 | 12 |
| Garoua | 54 | 162 | 12 |
| Kumba | 56 | 168 | 12 |
| Maroua | 74 | 222 | 12 |
| Ngaoundere | 41 | 123 | 12 |
| Yaounde | 224 | 1344 | 12 |
| Conakry | 105 | 420 | 12 |
| Bamako | 291 | 1164 | 12 |
| Niamey | 193 | 772 | 12 |
| Aba Ng | 51 | 153 | 12 |
| Benin City | 149 | 596 | 12 |
| Ibadan | 152 | 912 | 12 |
| Ilorin | 74 | 222 | 12 |
| Jos | 79 | 237 | 12 |
| Kano | 429 | 2574 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Maiduguri | 200 | 800 | 12 |
| Onitsha | 210 | 840 | 12 |
| Port Harcourt | 200 | 800 | 12 |
| Uyo | 55 | 165 | 12 |
| Dakar | 250 | 1500 | 12 |
| Arish | 23 | 46 | 12 |
| Asyut | 76 | 228 | 12 |
| Beni Suef | 56 | 168 | 12 |
| Damanhur | 77 | 231 | 12 |
| Damietta | 102 | 306 | 12 |
| Fayoum | 85 | 255 | 12 |
| Hurghada | 51 | 102 | 12 |
| Ismailia | 71 | 213 | 12 |
| Kafr El Sheikh | 53 | 106 | 12 |
| Luxor | 81 | 243 | 12 |
| Mahalla | 61 | 183 | 12 |
| Mansoura Eg | 72 | 216 | 12 |
| Minya | 68 | 204 | 12 |
| Port Said | 49 | 147 | 12 |
| Qena | 81 | 243 | 12 |
| Sohag | 58 | 174 | 12 |
| Suez | 85 | 255 | 12 |
| Tanta | 97 | 291 | 12 |
| Zagazig | 57 | 171 | 12 |
| Amarah | 70 | 210 | 12 |
| Baqubah | 94 | 282 | 12 |
| Basra | 304 | 1824 | 12 |
| Diwaniyah | 78 | 234 | 12 |
| Duhok | 80 | 240 | 12 |
| Erbil | 171 | 684 | 12 |
| Fallujah | 77 | 231 | 12 |
| Hillah | 88 | 264 | 12 |
| Karbala | 193 | 772 | 12 |
| Kirkuk | 168 | 672 | 12 |
| Kut | 69 | 207 | 12 |
| Mosul | 221 | 884 | 12 |
| Najaf | 198 | 792 | 12 |
| Nasiriyah | 76 | 228 | 12 |
| Ramadi | 74 | 222 | 12 |
| Samawah | 77 | 231 | 12 |
| Sulaymaniyah | 124 | 496 | 12 |
| Amman | 372 | 2232 | 12 |
| Aqaba | 59 | 118 | 12 |
| Irbid | 74 | 222 | 12 |
| Zarqa | 98 | 294 | 12 |
| Beirut | 182 | 728 | 12 |
| Sidon | 51 | 102 | 12 |
| Tripoli Lb | 69 | 207 | 12 |
| Gaza City | 51 | 153 | 12 |
| Hebron | 87 | 261 | 12 |
| Nablus | 96 | 288 | 12 |
| Abha | 95 | 285 | 12 |
| Al Kharj | 88 | 264 | 12 |
| Buraidah | 97 | 291 | 12 |
| Dammam | 285 | 1140 | 12 |
| Hail | 89 | 267 | 12 |
| Hofuf | 100 | 300 | 12 |
| Jeddah | 365 | 2190 | 12 |
| Jizan | 78 | 234 | 12 |
| Khamis Mushait | 115 | 345 | 12 |
| Mecca | 234 | 936 | 12 |
| Medina | 196 | 784 | 12 |
| Najran | 83 | 249 | 12 |

| City | Provisional trainsets | Cars | Open gates |
|---|---|---|---|
| Tabuk | 87 | 261 | 12 |
| Taif | 85 | 255 | 12 |
| Aleppo | 224 | 896 | 12 |
| Damascus | 238 | 952 | 12 |
| Deir Ez Zor | 84 | 252 | 12 |
| Hama | 84 | 252 | 12 |
| Homs | 61 | 183 | 12 |
| Idlib | 54 | 108 | 12 |
| Latakia | 69 | 207 | 12 |
| Raqqa | 59 | 177 | 12 |
| Tartus | 46 | 92 | 12 |
| Aden | 74 | 222 | 12 |
| Dhamar | 58 | 116 | 12 |
| Hodeidah | 47 | 141 | 12 |
| Ibb | 79 | 237 | 12 |
| Lahij | 48 | 96 | 12 |
| Mukalla | 88 | 264 | 12 |
| Sanaa | 292 | 1752 | 12 |
| Sayun | 41 | 82 | 12 |
| Taiz | 78 | 234 | 12 |